CM-SIP — Question Bank

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CM-SIP Extensive Question Bank

This question bank follows the sequence and style of Appendix E in the IBF CM-SIP guide, but expands it substantially. It is organised by chapter and section, includes Chapters 1 and 12 even though Appendix E does not provide separate blocks for them, and adds calculations, suitability decisions and scenario questions.

Answers and rationales are hidden inside expandable sections; click **Show answer** after attempting each question.

For multiple-response questions, select every correct option. Unless stated otherwise, use a 360-day convention for non-Singapore-Government-Securities products and the formulas in Appendix D of `CMSIP_AI.md`. These are original revision questions, not a reproduction of the source's review-question set.

Chapter 1 — Overview of Derivatives

1.1 Introduction and 1.2 Underlying asset classes

1. [**Single choice**] Which statement best defines a derivative?

a. A deposit whose principal is always insured. b. A bond with no maturity. c. A security that always gives ownership of a company. d. A contract whose value is linked to an underlying asset, rate, index, event or outcome.

Answer and explanation

Answer: d. A derivative derives value from an underlying; ownership does not automatically pass to the investor.

2. [**Multiple response**] Which are common uses of derivatives?

a. Hedging an existing exposure. b. Speculating on direction or volatility. c. Arbitrage between mispriced markets. d. Repackaging cash flows or creating a synthetic position.

Answer and explanation

Answer: a, b, c and d. All four are core uses.

3. [**Single choice**] A Singapore importer payable in USD in three months is most directly exposed to:

a. USD/SGD currency risk. b. Equity dilution risk only. c. The risk that a call option will always be exercised. d. No risk until payment day.

Answer and explanation

Answer: a. A currency forward, future or option may hedge the USD payable.

4. [**Multiple response**] Which asset classes are discussed as derivative underlyings?

a. FX. b. Money-market/short-term rates. c. Fixed income and credit. d. Equities and commodities.

Answer and explanation

Answer: a, b, c and d. The guide also extends these to indices, baskets, funds and other references.

5. [**Single choice**] Why can an NDF be useful for a restricted currency?

a. It eliminates all FX risk. b. It is always exchange traded. c. It guarantees physical delivery of the restricted currency. d. It is normally cash-settled without exchanging the notional currency.

Answer and explanation

Answer: d. An NDF settles the difference between the contracted rate and the fixing/spot rate.

6. [**Single choice**] Which statement about a CDS is correct?

a. It transfers or references credit protection on a debt obligation or index. b. It is a form of equity dividend. c. It is a physical commodity storage contract. d. It always transfers legal ownership of a bond.

Answer and explanation

Answer: a. The protection buyer pays a spread and receives a contingent payment after a defined credit event.

7. [**Multiple response**] Commodity-derivative pricing can be affected by:

a. Storage and insurance cost. b. Transport and delivery location. c. Seasonality, weather and supply/demand. d. The USD and interest rates.

Answer and explanation

Answer: a, b, c and d. These factors influence carry, convenience yield and spot/futures relationships.

1.3 Exchange-traded derivatives and 1.4 OTC derivatives

8. [**Multiple response**] Typical exchange-traded derivative features include:

a. Standardised contract terms. b. Central clearing. c. Daily mark-to-market/margining. d. Bespoke bilateral terms for every customer.

Answer and explanation

Answer: a, b and c. Bespoke terms are primarily an OTC feature.

9. [**Single choice**] Central clearing mainly reduces:

a. The need to post margin. b. Basis risk. c. All market losses. d. Bilateral counterparty exposure through the clearing mechanism.

Answer and explanation

Answer: d. A CCP becomes the buyer to every seller and seller to every buyer, but market, margin, liquidity and clearing risks remain.

10. [**Multiple response**] OTC customisation can increase exposure to:

a. Counterparty risk. b. Liquidity and close-out risk. c. Model, valuation and documentation risk. d. Legal and operational risk.

Answer and explanation

Answer: a, b, c and d. Customisation is valuable but less standardised and often less liquid.

11. [**Single choice**] Which post-crisis reform is consistent with the guide?

a. Move suitable standardised OTC derivatives toward electronic/exchange trading, CCP clearing and trade reporting. b. Remove capital requirements for uncleared derivatives. c. Replace all OTC products with physical shares. d. Ban all derivatives.

Answer and explanation

Answer: a. These are examples of the G20 reform direction described in the source.

1.5 Structured products and 1.6 Singapore market

12. [**Single choice**] A structured product should be analysed as:

a. Direct ownership of every underlying asset. b. An instrument with no issuer risk. c. A guaranteed deposit because it has a high coupon. d. A wrapper containing a principal component plus one or more derivative exposures.

Answer and explanation

Answer: d. The headline return can compensate the investor for selling protection, accepting conversion or bearing issuer/illiquidity risk.

13. [**Multiple response**] Before recommending a structured product, a representative should examine:

a. Trigger/barrier and payoff formula. b. Issuer/SPV and counterparty. c. Liquidity, early termination and fees. d. Currency, maximum loss and suitability.

Answer and explanation

Answer: a, b, c and d. A product label is not a complete risk description.

14. [**Single choice**] Which venues are identified in the guide's Singapore-market overview?

a. SGX-DT, ICE Futures Singapore and APEX. b. Only an unregulated bilateral telephone market. c. Only US exchanges. d. None; Singapore has no derivatives market.

Answer and explanation

Answer: a. The source also discusses Singapore's Asian equity-index, commodity, FX and rates role.

15. [**Scenario**] A distributor relies on a foreign target-market document and does not perform Singapore KYC/knowledge/suitability checks. What is the best conclusion?

a. Only the client's net worth matters. b. No checks are needed for exchange-traded products. c. This is acceptable because foreign product governance replaces local obligations. d. Product governance, disclosure, KYC, suitability, conflicts and local regulatory obligations still need to be addressed.

Answer and explanation

Answer: d. The MiFID II discussion in the source is an application example; it does not displace Singapore requirements.

Chapter 2 — Futures, Forwards and Swaps

2.1 Introduction and 2.2 General features

16. [**Single choice**] Which is the clearest difference between a future and a forward?

a. Futures are usually standardised/exchange cleared; forwards are usually customised/OTC. b. Futures have no maturity; forwards always expire daily. c. Forwards are always safer because they are bilateral. d. Futures cannot be closed before maturity.

Answer and explanation

Answer: a. Futures also use daily MTM/margin, while forwards generally have bilateral credit exposure and less secondary liquidity.

17. [**Multiple response**] Futures contracts commonly specify:

a. Contract size/multiplier. b. Tick size and price limits. c. Trading hours and last trading day. d. Settlement and delivery terms.

Answer and explanation

Answer: a, b, c and d. These define the tradable contract.

18. [**Calculation**] An E-mini contract is quoted at 5,000 and has a multiplier of 50. What is the notional value?

a. USD5,050. b. USD25,000. c. USD250,000. d. USD500,000.

Answer and explanation

Answer: c. `5,000 × 50 = USD250,000` per contract.

19. [**Calculation**] A Nikkei contract is at 22,500 with a JPY500 multiplier. Initial margin is 6.5%. What margin is required?

a. JPY1,462,500. b. JPY11,250,000. c. JPY73,125. d. JPY731,250.

Answer and explanation

Answer: d. Notional = JPY11,250,000; `11,250,000 × 6.5% = JPY731,250`.

20. [**Single choice**] Daily mark-to-market means:

a. Gains/losses are settled periodically based on the contract's changing market value. b. The investor pays the entire notional on day one. c. The underlying is always physically delivered daily. d. Margin is never adjusted.

Answer and explanation

Answer: a. Daily settlement creates cash-flow and margin-call risk.

2.3 Types and 2.4 settlement/clearing

21. [**Single choice**] A short-term interest-rate futures price is quoted as `100 − R`. If implied rate rises, the futures price generally:

a. Rises. b. Falls. c. Is unchanged. d. Becomes the spot equity index.

Answer and explanation

Answer: b. The quote is inversely related to the implied rate.

22. [**Calculation**] A short-rate future is quoted at 98.25. What implied rate does the source convention suggest?

a. 98.25%. b. 101.75%. c. 0.175%. d. 1.75%.

Answer and explanation

Answer: d. `R = 100 − 98.25 = 1.75%`.

23. [**Calculation**] A GBP/USD spot rate is 1.2422. UK rate is 0.70444%, US rate is 1.799%, and tenor is one year. Using interest-rate parity with USD as base, the theoretical rate is closest to:

a. 1.2557. b. 3.1743. c. 0.9989. d. 1.2487.

Answer and explanation

Answer: a. `F = 1.2422 × (1.0070444)/(1.01799)  1.2557` under the guide's quotation convention.

24. [**Multiple response**] A futures position can be resolved by:

a. Offset/close-out. b. Physical delivery where specified. c. Cash settlement where specified. d. Rolling into a later contract.

Answer and explanation

Answer: a, b, c and d. The appropriate route depends on the contract and investor's objective.

25. [**Single choice**] What does an SGX clearing house do?

a. Sets all market prices by itself. b. Clears/margins contracts and manages default processes as the central counterparty. c. Guarantees profitable trades. d. Removes the need to understand contract specifications.

Answer and explanation

Answer: b. Clearing reduces bilateral credit exposure but does not guarantee profit or eliminate market risk.

2.5 Pricing and 2.6 basis

26. [**Calculation**] Spot gold is 1,300, carry is 0.28% for the period and there is no income. Approximate fair futures value under `F=S(1+C)` is:

a. 1,328.00. b. 1,296.36. c. 1,300.00. d. 1,303.64.

Answer and explanation

Answer: d. `1,300 × 1.0028 = 1,303.64`.

27. [**Single choice**] In a cash-and-carry arbitrage when futures is too high, the trader normally:

a. Buys/finances the spot asset and sells futures. b. Sells spot and sells futures. c. Buys both spot and futures. d. Does nothing because price differences are always risk-free.

Answer and explanation

Answer: a. The spot position carries the asset into delivery while the short futures locks the sale price.

28. [**Multiple response**] Basis can be affected by:

a. Carry, storage, insurance and convenience yield. b. Time to expiry and yield-curve changes. c. Liquidity and contract/asset mismatch. d. Administered versus market rates and expectations.

Answer and explanation

Answer: a, b, c and d. Basis is not simply "the forecast price."

29. [**Single choice**] Contango usually means:

a. There is no carrying cost. b. Futures price is above spot. c. Futures price is below spot. d. Spot and futures are identical at every date.

Answer and explanation

Answer: b. Backwardation is the opposite relationship, though the exact level also reflects income and convenience yield.

30. [**Scenario**] A hedge is closed two months before expiry and the asset being hedged is similar but not identical to the futures underlying. The main residual risk is:

a. Voting-rights risk only. b. No risk because futures are cleared. c. Deposit-insurance risk. d. Basis risk.

Answer and explanation

Answer: d. Mismatch and early close-out prevent perfect convergence.

2.7 Swaps and 2.8 summary

31. [**Single choice**] In a vanilla interest-rate swap, the notional is normally:

a. Used to calculate interest cash flows but not exchanged. b. Always converted into shares. c. The option premium only. d. Exchanged physically at each reset.

Answer and explanation

Answer: a. Fixed and floating interest amounts are netted/paid on the notional.

32. [**Multiple response**] Which swap descriptions are correct?

a. An FX swap has near and forward legs. b. A currency swap exchanges debt-service obligations in two currencies. c. An equity swap exchanges equity/index return for fixed or floating payments. d. A commodity swap can exchange fixed and floating commodity prices.

Answer and explanation

Answer: a, b, c and d. These are distinct applications of OTC cash-flow exchange.

33. [**Single choice**] A CDS protection buyer generally:

a. Has no exposure to the protection seller. b. Pays a spread and receives protection after a defined credit event. c. Receives a fixed dividend and gives voting rights. d. Buys physical gold.

Answer and explanation

Answer: b. The protection seller's ability to pay is itself a counterparty risk.

34. [**Scenario**] A borrower pays floating interest but wants fixed cash flows. The most direct swap action is to:

a. Pay floating and receive floating. b. Buy a put warrant. c. Sell a commodity future. d. Pay fixed and receive floating.

Answer and explanation

Answer: d. The received floating leg offsets the borrower's floating liability.

35. [**Multiple response**] Why can a swap still be risky after it achieves the desired hedge?

a. Counterparty/default risk. b. Basis, reset and mismatch risk. c. Collateral, legal and documentation risk. d. Early termination/valuation and liquidity risk.

Answer and explanation

Answer: a, b, c and d. Hedging reduces a targeted risk, not every risk.

Chapter 3 — Futures Strategies

3.1 Participants and 3.2 SGX orders

36. [**Single choice**] A portfolio manager sells index futures while continuing to hold the cash portfolio. The manager is acting primarily as a:

a. Hedger. b. Market maker. c. Arbitrageur with no cash exposure. d. Depositor.

Answer and explanation

Answer: a. The futures position offsets the portfolio's market exposure.

37. [**Multiple response**] Which participants can be active in futures markets?

a. Hedgers. b. Speculators and hedge funds. c. Market makers and proprietary firms. d. Arbitrageurs, portfolio managers and official institutions.

Answer and explanation

Answer: a, b, c and d. Different participants supply liquidity and transfer/accept risk.

38. [**Single choice**] A buy limit order at $10.00 when the ask is $10.02 will generally:

a. Execute immediately at $10.02. b. Rest in the queue at $10.00 or better until filled, amended or cancelled. c. Become a sell order. d. Guarantee execution at the close.

Answer and explanation

Answer: b. A limit order controls price but not execution.

39. [**Single choice**] In an MTL buy order, any unfilled quantity normally:

a. Converts into an option. b. Is physically delivered. c. Is cancelled immediately in all cases. d. Remains at the price where the filled portion executed, subject to the rules.

Answer and explanation

Answer: d. MTL provides market execution for the first fill and a limit-like resting balance.

40. [**Multiple response**] A stop order may be used to:

a. Cut a losing long position. b. Protect profit using a trailing stop. c. Enter after an upside breakout. d. Enter short after a downside breakdown.

Answer and explanation

Answer: a, b, c and d. The trigger direction determines the resulting buy/sell order.

3.3 Outright trades and 3.4 spreads

41. [**Single choice**] Which is an outright futures trade?

a. Buy one contract to express a directional view. b. Buy near month and sell deferred month. c. Buy one call and sell another call. d. Buy one contract and sell a related contract simultaneously.

Answer and explanation

Answer: a. The other choices are spread or options structures.

42. [**Multiple response**] Spread-trade risks include:

a. Correlation breakdown. b. One leg filling before the other. c. Liquidity and margin changes. d. The spread moving in the wrong direction.

Answer and explanation

Answer: a, b, c and d. Lower spread margin does not mean no risk.

43. [**Single choice**] A calendar spread usually involves:

a. Two unrelated currencies only. b. Same underlying with different delivery months. c. Different underlyings and same maturity only. d. A cash share and a bank deposit.

Answer and explanation

Answer: b. It is also called a horizontal/time spread.

44. [**Scenario**] An investor expects the near-month price to strengthen relative to the deferred-month price. The natural calendar-spread expression is:

a. Short near and long deferred. b. Long both months. c. Short both months. d. Long near and short deferred.

Answer and explanation

Answer: d. Profit depends on the near-minus-deferred spread widening.

45. [**Single choice**] A butterfly futures spread commonly has which weights across three maturities?

a. `+1, −2, +1`. b. `+1, +1, +1`. c. `+1, −1` only. d. `+2, +2, −1`.

Answer and explanation

Answer: a. The structure targets curvature rather than a simple outright direction.

46. [**Single choice**] Compared with a butterfly, a condor normally has:

a. Only one maturity. b. Four different maturities and a wider but lower peak profit zone. c. Two legs and unlimited loss. d. No sensitivity to volatility.

Answer and explanation

Answer: b. A condor uses four different delivery points.

47. [**Single choice**] The TED spread is associated with:

a. Gold versus silver storage only. b. Two equity dividends. c. A company's ordinary share split. d. T-bill/Treasury versus Eurodollar/LIBOR pricing and credit/liquidity conditions.

Answer and explanation

Answer: d. It is not a pure directionless interest-rate measure.

3.5 Hedging and portfolio management

48. [**Single choice**] The purpose of hedging is best described as:

a. Exchange uncertain exposure for a more predictable outcome, usually leaving basis risk. b. Guarantee a higher return than speculation. c. Avoid all margin. d. Eliminate every possible loss.

Answer and explanation

Answer: a. A hedge targets a defined risk and may introduce other risks/costs.

49. [**Multiple response**] A hedge programme should specify:

a. Exposure and objective. b. Correlation/hedgeability and vehicle. c. Horizon, target ratio and roll/close-out. d. Monitoring, limits and residual risk.

Answer and explanation

Answer: a, b, c and d. A hedge is a managed programme, not a single trade.

50. [**Calculation**] A cash exposure is \$100m, a futures contract is \$1m and the one-to-one hedge ratio is 1. How many contracts are required for a complete hedge?

a. 10. b. 50. c. 100. d. 1,000.

Answer and explanation

Answer: c. `100m/1m = 100`; a short position would normally hedge a long cash exposure.

51. [**Calculation**] A portfolio is SGD5.1025m, the futures quote is 4,100, value per point is SGD50 and beta is 1.2. Approximate contracts using `N=VP/(F×T)×` .

a. 15. b. 30. c. 60. d. 120.

Answer and explanation

Answer: b. `5,102,500/(4,100×50)×1.2  29.77`, rounded to about 30.

52. [**Single choice**] A long-term bond hedge should primarily compare:

a. Face value only. b. PVBP/DV01 and the conversion factor of the deliverable futures bond. c. The issuer's dividend yield. d. Equity beta only.

Answer and explanation

Answer: b. Dollar interest-rate sensitivity is the relevant exposure.

53. [**Multiple response**] Futures can support portfolio techniques such as:

a. Asset allocation and rebalancing. b. Synthetic exposure and cash equitisation. c. Duration or equity overlay hedging. d. Temporary market exposure while cash is invested.

Answer and explanation

Answer: a, b, c and d. The efficiencies come with basis, roll, margin and tracking risks.

3.6 Portfolio techniques and 3.7 arbitrage

54. [**Single choice**] Cash-and-futures arbitrage is most credible when:

a. The asset cannot be bought or borrowed. b. There is no way to close either leg. c. The apparent spread exceeds funding, trading, storage, margin and delivery costs. d. The spread is smaller than all transaction costs.

Answer and explanation

Answer: c. The theoretical arbitrage must survive real-world costs and constraints.

55. [**Multiple response**] Why might an apparent futures arbitrage not be risk-free?

a. Funding/borrow constraints. b. Basis and value-date mismatch. c. Liquidity, bid–ask and margin. d. Delivery options or counterparty issues.

Answer and explanation

Answer: a, b, c and d. The textbook cash-and-carry assumptions are often violated.

56. [**Scenario**] A trader buys the cash asset, sells an overpriced future and plans to deliver at maturity. The position is a:

a. Reverse cash-and-carry. b. Bull put spread. c. Covered call. d. Cash-and-carry arbitrage.

Answer and explanation

Answer: d. The physical/cash asset is carried against a futures sale.

57. [**Single choice**] A strip differs from a stack because a strip generally:

a. Uses sequential futures delivery months corresponding to the exposure periods. b. Uses only the nearest contract and never rolls. c. Is always an option. d. Has no basis risk.

Answer and explanation

Answer: a. A stack concentrates in one/deferred contract and may be rolled to match a loan's reset dates.

58. [**Scenario**] A company has a floating-rate loan and expects rates to rise. A short Eurodollar/short-rate futures hedge aims to:

a. Remove all credit risk from the borrower. b. Gain as futures prices fall when rates rise, offsetting higher loan interest. c. Increase its floating-rate exposure. d. Receive dividends.

Answer and explanation

Answer: b. The hedge offsets rate sensitivity, subject to correlation and basis.

59. [**Multiple response**] A 130/30 equity strategy means:

a. Long 130% of portfolio value. b. Short 30%. c. Net 100% equity exposure before other effects. d. No leverage or short-sale risk.

Answer and explanation

Answer: a, b and c. It still carries leverage, borrow, stock-selection and market risks.

60. [**Single choice**] A player who sells contracts while holding the underlying stock is most likely:

a. A market maker with no risk. b. A depositor. c. A call-option buyer. d. A hedger.

Answer and explanation

Answer: d. The futures sale reduces cash-market exposure.

Chapter 4 — Options

4.1–4.2 Basics and definitions

61. [**Single choice**] The buyer of a call option has:

a. No premium obligation. b. The obligation to buy. c. The right, but not the obligation, to buy at the strike. d. The right to sell only.

Answer and explanation

Answer: c. The buyer pays premium for optionality; the writer has the contingent obligation.

62. [**Single choice**] An American option may be exercised:

a. Only if it is OTM. b. Only by the writer. c. Only at expiry. d. Any time up to expiry, subject to its terms.

Answer and explanation

Answer: d. European exercise is expiry-only.

63. [**Calculation**] A call has strike \$50, underlying \$54 and premium \$6. Its intrinsic value and time value are:

a. \$4 and \$2. b. \$50 and \$4. c. \$54 and \$6. d. \$6 and \$0.

Answer and explanation

Answer: a. Call intrinsic = `54-50=$4`; time value = `$6-$4=$2`.

64. [**Multiple response**] For a long call, which changes generally increase value, all else equal?

a. Higher underlying price. b. Higher volatility. c. More time to expiry. d. Higher dividends.

Answer and explanation

Answer: a, b and c. Higher dividends generally reduce call value.

4.3–4.6 Payoff, factors and parity

65. [**Calculation**] A long call has strike \$100 and premium \$7. What is breakeven at expiry?

a. \$100. b. \$107. c. Unlimited. d. \$93.

Answer and explanation

Answer: b. Breakeven = `X + premium = $107`.

66. [**Single choice**] Maximum loss for a long put is normally:

a. Zero only if the stock rises. b. Unlimited. c. The premium paid. d. The strike plus premium.

Answer and explanation

Answer: c. The put may expire worthless, but the holder's loss is limited to premium.

67. [**Single choice**] A naked short call has:

a. No obligation. b. A guaranteed maximum loss equal to premium. c. A right to buy the underlying. d. Unlimited theoretical loss if the underlying rises sharply.

Answer and explanation

Answer: d. The writer must deliver/satisfy the call if exercised.

68. [**Single choice**] Put-call parity for European options is:

a. `C + PV(X) = P + S`. b. `C + P = S - PV(X)`. c. `C - PV(X) = P + S`. d. `C + S = P - PV(X)`.

Answer and explanation

Answer: a. Same underlying, strike, expiry, quantity and consistent dividend assumptions are required.

69. [**Multiple response**] Under parity, a synthetic long stock can be created by:

a. A bond alone. b. Long call and short put, with the same strike/expiry. c. Long put and short call only. d. Two long calls with different strikes.

Answer and explanation

Answer: b. Long call − short put plus the parity bond/financing relationship reproduces the stock cash flows.

70. [**Scenario**] A trader buys a call and a put at the same strike and expiry, expecting a large move but no direction. This is a:

a. Bull put spread. b. Covered call. c. Long straddle. d. Short strangle.

Answer and explanation

Answer: c. A straddle is long call and long put at the same strike.

4.7–4.10 Valuation, Greeks and other underlyings

71. [**Multiple response**] Which statements about Greeks are correct?

a. Delta measures option-price sensitivity to the underlying. b. Gamma measures the change in delta. c. Theta measures time decay. d. Vega measures volatility sensitivity.

Answer and explanation

Answer: a, b, c and d. Rho measures sensitivity to interest rates.

72. [**Single choice**] Gamma is often largest when an option is:

a. Certain to expire worthless immediately. b. A zero-coupon bond. c. Deep ITM and years from expiry. d. Near ATM and close to expiry.

Answer and explanation

Answer: d. Delta changes rapidly near the strike as expiry approaches.

73. [**Single choice**] Vega is generally positive for:

a. Long calls and long puts. b. Short calls only. c. Long stock only. d. Short straddles only.

Answer and explanation

Answer: a. Higher implied volatility generally increases the value of long options.

74. [**Multiple response**] Equity-index option analysis should consider:

a. Cash settlement and multiplier. b. Index composition and liquidity. c. Basis/correlation with a portfolio. d. OTC counterparty risk where applicable.

Answer and explanation

Answer: a, b, c and d. An index is not automatically a perfect hedge.

75. [**Single choice**] A payer swaption generally benefits its buyer when:

a. A commodity is delivered. b. Rates rise above the fixed strike. c. Rates fall to zero only. d. Equity dividends rise.

Answer and explanation

Answer: b. The buyer gains the right to enter a pay-fixed/receive-floating swap at the strike.

4.11–4.14 Strategies and hedging

76. [**Multiple response**] A bull call spread normally consists of:

a. Long lower-strike call. b. Short higher-strike call. c. Same expiry/underlying. d. Unlimited loss in all cases.

Answer and explanation

Answer: a, b and c. The short call caps gain and helps fund the debit; loss is generally limited to net premium.

77. [**Single choice**] A bear put spread profits primarily when:

a. The underlying rises without limit. b. Interest rates fall only. c. The underlying falls. d. Volatility is always zero.

Answer and explanation

Answer: c. It is a debit bearish structure with defined risk and capped profit.

78. [**Single choice**] A long strangle is generally cheaper than a long straddle because:

a. It contains only one option. b. It uses OTM options at different strikes and needs a larger move. c. It has no premium. d. It is always risk-free.

Answer and explanation

Answer: b. The trade-off is lower initial cost for wider breakevens.

79. [**Scenario**] An investor owns shares and wants to floor downside but retain upside above the strike. The most direct option hedge is:

a. Sell a naked put. b. Buy a short call only. c. Sell the shares and buy a bond. d. Buy a protective put.

Answer and explanation

Answer: d. A protective put is long stock plus long put.

80. [**Multiple response**] A zero-cost collar normally involves:

a. Buying a put. b. Selling an OTM call. c. Premiums approximately offsetting. d. Unlimited upside after the call strike.

Answer and explanation

Answer: a, b and c. The sold call limits upside; the structure is not free of risk.

Chapter 5 — Structured Warrants and DLCs

5.1–5.4 Warrants, terms and price factors

81. [**Single choice**] A structured warrant is normally:

a. Issued by a third-party financial institution and linked to an underlying. b. A direct share with voting rights. c. A bank deposit with guaranteed return. d. A futures contract with daily margin only.

Answer and explanation

Answer: a. Issuer credit risk is therefore central.

82. [**Calculation**] A call warrant has `S=$2.40`, `X=$0.80`, price `$0.40`, and conversion ratio `n=1`. What is gearing?

a. 2×. b. 6×. c. 8×. d. 0.67×.

Answer and explanation

Answer: b. `2.40/(0.40×1)=6×`.

83. [**Calculation**] A call warrant has strike $1.00, price $1.20 and `n=1`. What is conversion price?

a. $1.00. b. $1.20. c. $2.20. d. $0.20.

Answer and explanation

Answer: c. `X+nWP = 1+1.20 = $2.20`.

84. [**Calculation**] A put warrant has `S=$1.50`, `X=$2.00`, `n=2`. What is intrinsic value per warrant?

a. $0.50. b. $1.00. c. $2.50. d. $0.25.

Answer and explanation

Answer: d. `(2.00−1.50)/2 = $0.25`.

85. [**Multiple response**] Warrant value can be affected by:

a. Underlying price and strike. b. Time and implied volatility. c. Dividends and interest rates. d. Conversion ratio, liquidity and issuer credit.

Answer and explanation

Answer: a, b, c and d. Gearing alone is not a complete valuation.

5.5–5.11 Trading, valuation, DMM, settlement and exotic warrants

86. [**Single choice**] The maximum loss for a buyer of a warrant, excluding transaction costs and issuer default mechanics, is generally:

a. The amount paid for the warrant. b. Unlimited. c. The underlying share price plus strike. d. The conversion ratio.

Answer and explanation

Answer: a. The warrant may expire worthless.

87. [**Single choice**] A DMM's role is primarily to:

a. Set the underlying company's dividend. b. Provide quotes/liquidity subject to exchange/product conditions. c. Guarantee an investor's profit. d. Remove issuer risk.

Answer and explanation

Answer: b. Quote obligations can be suspended or limited during disruptions.

88. [**Multiple response**] Corporate actions may cause a warrant issuer to adjust:

a. Strike price. b. Conversion ratio. c. Cash settlement/final reference mechanics. d. The investor's ordinary voting rights.

Answer and explanation

Answer: a, b and c. A warrant holder does not become an ordinary shareholder merely by holding the warrant.

89. [**Single choice**] A five-day average settlement reference means the investor should not assume:

a. The issuer has credit risk. b. A put has downside exposure. c. The final payoff uses one expiry-day closing print. d. The warrant has any time value.

Answer and explanation

Answer: c. The product's settlement observation must be read precisely.

90. [**Multiple response**] The guide's exotic warrant examples include:

a. Index and basket warrants. b. Currency-translated and FX warrants. c. Commodity warrants. d. Discount/capped or yield-enhanced warrants.

Answer and explanation

Answer: a, b, c and d. Each adds underlying, correlation, currency or capped-upside issues.

5.12 DLCs and 5.13 summary

91. [**Single choice**] A 3× daily long DLC is designed to target approximately:

a. Three times the underlying's total return over any five-year period. b. A fixed 3% coupon. c. No loss if the underlying falls. d. Three times the underlying's daily percentage move before fees/adjustments.

Answer and explanation

Answer: d. Daily reset and compounding make long-horizon results path dependent.

92. [**Scenario**] An index rises 10% one day and falls 9.09% the next day, approximately returning to its starting level. A daily 3× long product may:

a. Suffer path-dependent loss because daily returns compound. b. Pay a fixed coupon. c. Convert into ordinary shares automatically. d. Return exactly zero before costs.

Answer and explanation

Answer: a. Daily leverage is not the same as a simple multiple of total-period return.

93. [**Multiple response**] DLC risks include:

a. Daily reset/compounding. b. Airbag/observation and suspension mechanics. c. Issuer, DMM, liquidity and fee risk. d. Large losses when the underlying moves against the position.

Answer and explanation

Answer: a, b, c and d. The source describes them as short-term, high-risk products.

94. [**Single choice**] If a 5× DLC has an adverse airbag threshold of 10%, the investor should:

a. Assume the product cannot suspend. b. Read the term sheet for the trigger, observation level, suspension and reset mechanics. c. Treat it as a deposit. d. Ignore the underlying's path.

Answer and explanation

Answer: b. The exact terms control; the guide gives illustrative thresholds, not universal rules.

Chapter 6 — Barrier Options, Binary Options and CBBCs

6.1–6.5 Barriers

95. [**Single choice**] For a single up-and-out call, the barrier is normally:

a. Below zero. b. Below spot. c. Above spot. d. Always equal to strike.

Answer and explanation

Answer: c. If spot touches/crosses the upper barrier, the call may knock out.

96. [**Single choice**] A knock-in option becomes active:

a. Immediately when purchased. b. Only when the holder sells it. c. Only if the issuer defaults. d. Only after the defined barrier event occurs.

Answer and explanation

Answer: d. Before activation, the option may have no vanilla payoff.

97. [**Multiple response**] A barrier option's outcome can depend on:

a. Whether the barrier was ever touched. b. Continuous versus discrete monitoring. c. Rebate and settlement terms. d. Gaps, volatility, issuer and liquidity.

Answer and explanation

Answer: a, b, c and d. The same expiry price can produce different outcomes depending on path.

98. [**Scenario**] An investor buys a down-and-out put and the underlying gaps below the barrier overnight. The main risk is:

a. The option may be knocked out and the premium lost despite the eventual fall. b. The investor automatically receives unlimited profit. c. The barrier becomes irrelevant. d. The put becomes a call.

Answer and explanation

Answer: a. Barrier knock-out is path-dependent and can occur during a gap.

99. [**Multiple response**] Barrier applications in the guide include:

a. Shark's-fin/barrier capital-preservation structures. b. Barrier reverse convertibles. c. Bonus certificates. d. Barrier straddles.

Answer and explanation

Answer: a, b, c and d. Their risk-return trade-offs differ by barrier direction and embedded leg.

6.6–6.9 Binary, basket and knock-out warrants

100. [**Single choice**] A one-touch binary generally pays when:

a. The investor pays no premium. b. The barrier is touched under the stated monitoring rules. c. The underlying finishes exactly at zero. d. The investor owns the physical asset.

Answer and explanation

Answer: b. A no-touch pays under the opposite path condition.

101. [**Multiple response**] A basket option adds risk from:

a. Correlation. b. Constituent weights and rebalancing. c. The weakest/worst-of asset where applicable. d. Dividend and liquidity differences.

Answer and explanation

Answer: a, b, c and d. Basket payoff is not simply the average of single-name outcomes.

102. [**Single choice**] A knock-out warrant differs from a plain warrant mainly because:

a. It is always a deposit. b. It has no underlying. c. A barrier can terminate it before ordinary expiry. d. It cannot be issued by a bank.

Answer and explanation

Answer: c. The barrier creates early-termination risk.

6.10–6.13 CBBCs and comparison

103. [**Single choice**] A bull CBBC benefits primarily when the underlying:

a. Falls through the call price. b. Stops trading permanently. c. Pays no dividend. d. Rises.

Answer and explanation

Answer: d. A fall to/below the bull call price can cause an MCE.

104. [**Single choice**] For a bear CBBC, an MCE generally occurs when the underlying:

a. Rises to or above the call price. b. Falls below the strike only. c. Pays a dividend. d. Remains unchanged.

Answer and explanation

Answer: a. A bear profits from falls but is terminated by an adverse rise.

105. [**Multiple response**] Which statements are correct about CBBC categories?

a. Category N has no residual value after an MCE under the source's description. b. Category R may have residual value, which may still be zero. c. MCE creates early termination. d. Category R is a guarantee of positive residual value.

Answer and explanation

Answer: a, b and c. Category R does not guarantee a positive residual.

106. [**Calculation**] A bull CBBC has underlying 1,050, strike 1,000, financial cost 10 and conversion ratio 10. Simplified theoretical price is:

a. 4. b. 5. c. 6. d. 10.

Answer and explanation

Answer: c. `[(1,050−1,000)+10]/10 = 6`.

107. [**Calculation**] A CBBC underlying is 1,200, price is \$0.60 and conversion ratio is 2,000. Using `underlying/(CBBC price × conversion ratio)`, gearing is:

a. 2,000×. b. 1×. c. 2×. d. 1,000×.

Answer and explanation

Answer: b. `1,200/(0.60×2,000)=1×`. The ratio convention must always be confirmed in the term sheet.

108. [**Multiple response**] CBBC risks include:

a. MCE/gap and residual-value risk. b. Issuer and liquidity/DMM risk. c. Leverage and underlying suspension. d. Currency, financing and maturity risk.

Answer and explanation

Answer: a, b, c and d. "No margin" does not mean "low risk."

Chapter 7 — Structured Deposits and Other Structured Products

7.1–7.5 Suitability, deposits and strategies

109. [**Multiple response**] A suitability review for a structured product should include:

a. Risk appetite and knowledge. b. Liquidity needs and investment horizon. c. Currency, restrictions and concentration. d. Principal-loss, counterparty, leverage and early-exit risks.

Answer and explanation

Answer: a, b, c and d. All affect whether the product fits the client.

110. [**Single choice**] In the guide's Singapore description, a structured deposit is:

a. An unsecured debenture rather than a deposit. b. A direct shareholding. c. A bank deposit with a structured/linked return and principal paid in full at maturity, subject to the product/issuer terms. d. Always deposit-insured.

Answer and explanation

Answer: c. The source specifically cautions that it is not deposit-insured and early withdrawal may lose principal/return.

111. [**Scenario**] A client wants a higher USD yield and is willing to receive SGD if USD strengthens beyond a strike. A DCI is suitable only if the client:

a. Believes a high rate eliminates FX risk. b. Requires guaranteed USD principal in all outcomes. c. Cannot tolerate alternate-currency ownership. d. Understands and accepts conversion and currency loss.

Answer and explanation

Answer: d. DCI yield compensates for the short conversion option.

112. [**Multiple response**] Structured deposits may include:

a. Knock-out/knock-in features. b. Bull call or put spreads. c. One-touch and range/double-no-touch features. d. Accumulator/decumulator mechanics.

Answer and explanation

Answer: a, b, c and d. Each changes the conversion, yield or purchase obligation.

113. [**Single choice**] An accumulator investor should be prepared to:

a. Buy the reference stock at the strike even when market price is below it. b. Receive unlimited upside without obligations. c. Sell the stock only when the bank asks. d. Avoid all margin calls.

Answer and explanation

Answer: a. The discount is compensation for the short-put/purchase obligation.

114. [**Single choice**] A 1×2 accumulator generally means the investor may buy:

a. Only one share over the entire tenor. b. Twice the predefined quantity when the price is below strike, subject to terms. c. No shares below strike. d. Two different currencies at a fixed rate.

Answer and explanation

Answer: b. Gearing magnifies downside quantity.

Chapter 8 — Structured Notes

8.1–8.3 Characteristics and types

115. [**Single choice**] A structured note is best described as:

a. A guaranteed deposit with no issuer risk. b. A futures exchange membership. c. Debt with an embedded derivative whose return is linked to an underlying. d. Direct ownership of the underlying.

Answer and explanation

Answer: c. The investor's claim is normally on the issuer/SPV.

116. [**Multiple response**] Structured-note underlyings can include:

a. Interest rates. b. Equities and indices. c. Currencies and commodities. d. Credit references and baskets.

Answer and explanation

Answer: a, b, c and d. The wrapper does not limit the derivative reference.

117. [**Single choice**] A note issued through an SPV may expose the investor to:

a. No credit risk because it is off balance sheet. b. Only ordinary-share voting risk. c. Guaranteed access to underlying assets. d. The SPV and possibly swap/issuer/underlying-credit risk described in the documents.

Answer and explanation

Answer: d. Off-balance-sheet status does not itself guarantee parent support or asset recourse.

8.4–8.6 RAN, callable notes, ELN and CLN

118. [**Calculation**] A RAN pays 6% for 60 of 90 observation days and 1% for the remaining 30 days. Weighted annualised payout is:

a. 4.33%. b. 6.00%. c. 1.00%. d. 3.00%.

Answer and explanation

Answer: a. `6%×60/90 + 1%×30/90 = 4.33%`.

119. [**Single choice**] A callable note creates reinvestment risk because:

a. The note becomes a share automatically. b. The issuer may redeem when rates are favourable to the issuer, leaving the investor to reinvest at lower rates. c. The investor can never receive principal. d. Calls always occur when rates rise.

Answer and explanation

Answer: b. The issuer, not the investor, generally controls the call.

120. [**Calculation**] An inverse floater has `X=8%`, leverage 2 and floating index 3%. Before any floor, coupon is:

a. 2%. b. 5%. c. 6%. d. 14%.

Answer and explanation

Answer: a. `8% − 2×3% = 2%`. A floor may prevent the coupon falling below the stated minimum, but before the floor the coupon is 2%.

121. [**Single choice**] A bull ELN most closely embeds:

a. A long futures contract only. b. A CDS protection purchase. c. A short put on the equity. d. A long put with unlimited protection.

Answer and explanation

Answer: c. Below strike, the investor may receive the falling shares/cash.

122. [**Multiple response**] A CLN investor may bear:

a. Issuer/SPV credit risk. b. Reference-entity credit-event risk. c. First-to-default correlation/concentration risk. d. Loss/reduced principal after a defined event.

Answer and explanation

Answer: a, b, c and d. The enhanced coupon is compensation for selling credit protection.

8.7–8.9 Market access and suitability

123. [**Single choice**] An ETN is generally:

a. A segregated pool that always owns all constituents. b. A deposit with deposit insurance. c. An ordinary share with voting rights. d. Unsecured listed debt tracking an index.

Answer and explanation

Answer: d. Issuer credit risk is central.

124. [**Multiple response**] Before buying a structured note, assess:

a. Liquidity/lock-up and early termination. b. Terms, fees and settlement. c. Issuer/SPV, collateral and recourse. d. Risk/return, currency and investor objective.

Answer and explanation

Answer: a, b, c and d. A high coupon alone is insufficient.

125. [**Single choice**] A PHS should prominently explain:

a. Key features, suitability, key risks and potential loss of principal. b. Only the issuer's marketing slogan. c. Only historical performance. d. No definitions or contact details.

Answer and explanation

Answer: a. The PHS exists to make the product understandable.

Chapter 9 — Structured Funds and ETFs

126. [**Single choice**] A structured fund is generally:

a. A single ordinary share. b. A CIS combining assets and/or derivatives to produce a targeted risk-return outcome. c. Always a direct bank deposit. d. Unregulated by any CIS framework.

Answer and explanation

Answer: b. Governance, custody, NAV and disclosure remain important.

127. [**Multiple response**] Fund performance/assessment can involve:

a. NAV and total return. b. TER/management and other costs. c. Sharpe ratio and tracking error. d. Underlying, share class and distribution terms.

Answer and explanation

Answer: a, b, c and d. No single ratio captures all fund risks.

128. [**Calculation**] A CPPI fund is worth \$100, floor \$85, crash size 20%. Risky allocation is:

a. \$15. b. \$25. c. \$75. d. \$85.

Answer and explanation

Answer: c. Multiplier = 5; cushion = 15; $risky =$‘ $5 \times 15 =$75‘.

129. [**Scenario**] In a prolonged range-bound market, CPPI may:

a. Guarantee full participation in every recovery. b. Remove floor/gap risk. c. Buy risky assets after rises and sell after falls. d. Always buy low and sell high.

Answer and explanation

Answer: c. Rebalancing can create whipsaw and cash-lock risk.

130. [**Multiple response**] ETF risks include:

a. Tracking error and NAV premium/discount. b. Bid–ask, liquidity, roll and cash drag. c. FX and counterparty/collateral risk where synthetic. d. Market, leverage/reset and closure risk.

Answer and explanation

Answer: a, b, c and d. Exchange trading does not eliminate product risk.

Chapter 10 — Contracts for Differences

10.1–10.5 Definition, characteristics and models

131. [**Single choice**] A CFD investor normally:

a. Receives a fixed deposit rate. b. Cannot take a short position. c. Owns the underlying share and receives voting rights. d. Gains/loses the cash difference between opening and closing values without physical ownership.

Answer and explanation

Answer: d. Dividend/corporate-action adjustments may apply, but ownership and voting rights do not.

132. [**Multiple response**] CFD features may include:

a. Leverage and margin. b. Long/short flexibility and cash settlement. c. Global underlying access. d. Financing, rollover and corporate-action adjustments.

Answer and explanation

Answer: a, b, c and d. Benefits and costs must be assessed together.

133. [**Single choice**] The source identifies which CFD model as predominant in Singapore?

a. Direct market access (DMA). b. Physical delivery only. c. An unlisted mutual fund. d. A gold-storage model.

Answer and explanation

Answer: a. Market-maker and exchange-traded models also exist.

134. [**Multiple response**] The CFD provider should make clear:

a. Initial/maintenance margin. b. Margin-call and liquidation procedures. c. Financing, commission, GST and rollover costs. d. RDS acknowledgement and client liability for shortfall.

Answer and explanation

Answer: a, b, c and d. The smaller initial outlay does not cap the loss.

10.6 Mechanics, orders and costs

135. [**Single choice**] A market order seeks:

a. A guaranteed price chosen by the investor. b. The best available bid/ask at execution. c. A price only after expiry. d. A physical share certificate.

Answer and explanation

Answer: b. Price can move before execution.

136. [**Single choice**] A limit buy at $10.00 when ask is $10.02 will normally:

a. Become a short sale. b. Execute at $10.02 immediately. c. Queue at $10.00 or lower. d. Trigger a stop at $9.50.

Answer and explanation

Answer: c. The price condition controls execution.

137. [**Scenario**] A long CFD stop-loss is set at $9.50, but the market gaps from $9.60 to $9.30. The investor should understand that:

a. The stop guarantees $9.50. b. The provider must pay the gap. c. The position becomes a long option. d. Execution may occur worse than $9.50 unless a guaranteed stop applies.

Answer and explanation

Answer: d. Stop orders are vulnerable to gaps and slippage.

138. [**Calculation**] An investor buys 10,000 CFDs at $2.00 and closes at $2.20. Commission is 0.4%, GST on commission 8%, financing 5% p.a. for 10 days. The source's approximate net profit is:

a. $1,790.78. b. $209.22. c. $86.40. d. $2,000.00.

Answer and explanation

Answer: a. Gross gain $2,000 less buy/sell costs $181.44 and financing $27.78 = $1,790.78.

139. [**Calculation**] A short CFD sells 10,000 at $3.30 and covers at $3.10. Under the source's 20% margin, 0.4% commission, 8% GST and five-day 6% financing example, ROI on margin is closest to:

a. 5.00%. b. 25.70%. c. 100%. d. −27.56%.

Answer and explanation

Answer: b. Net profit $1,696.02 divided by $6,600 margin 25.70%.

10.7–10.9 Global markets and strategies

140. [**Multiple response**] CFDs may reference:

a. Shares, indices and sectors. b. FX, commodities and government treasuries. c. ETFs, volatility and real estate. d. Only Singapore-listed common shares.

Answer and explanation

Answer: a, b and c. Availability depends on the provider.

141. [**Single choice**] Dividend-capture CFD traders generally seek shares with:

a. Guaranteed price appreciation. b. Only a stock split. c. Strong earnings and high payout potential. d. No dividends and no liquidity.

Answer and explanation

Answer: c. The price may not fall by the full dividend, but the trade remains risky and cost-sensitive.

142. [**Scenario**] A pairs trade is "market neutral" because:

a. It has no commissions. b. Both legs are always long. c. It is guaranteed profitable. d. Long and short related positions aim to offset broad market direction.

Answer and explanation

Answer: d. Correlation can break and both legs can lose.

143. [**Multiple response**] Pairs-trading risks include:

a. Relationships may not revert. b. One or both assets may move adversely. c. Double commissions/financing. d. Deal risk in merger arbitrage.

Answer and explanation

Answer: a, b, c and d. "Market neutral" does not mean risk-free.

144. [**Single choice**] Compared with equity futures, CFDs generally:

a. Have provider/OTC counterparty risk, can be rolled and explicitly charge financing. b. Have fixed exchange expiry and no financing. c. Receive no dividend adjustments. d. Cannot be short.

Answer and explanation

Answer: a. Equity futures are exchange-traded, fixed maturity, and financing is embedded in pricing.

Chapter 11 — Key Products and Investment Risks

11.1 Basic product risks

145. [**Multiple response**] Futures risks include:

a. Basis risk. b. Liquidity/lock-limit risk. c. Leverage and margin-call risk. d. Guaranteed loss limited to initial margin.

Answer and explanation

Answer: a, b and c. Loss can exceed initial margin.

146. [**Single choice**] Open interest helps assess futures:

a. Deposit-insurance coverage. b. Liquidity and the number of contracts still open. c. The issuer's dividend policy. d. The option's theta only.

Answer and explanation

Answer: b. Volume and open interest are useful liquidity indicators.

147. [**Multiple response**] Option holders may:

a. Lose the entire premium. b. Face time decay and illiquidity. c. Be affected by exercise terms and trading halts. d. Have unlimited losses as ordinary long-option holders.

Answer and explanation

Answer: a, b and c. A long option's loss is generally limited to premium, subject to costs/issuer failure.

148. [**Single choice**] Naked option writers are exposed to:

a. No exercise risk. b. Guaranteed profit. c. Potentially unlimited short-call loss and large short-put loss. d. Only the premium paid.

Answer and explanation

Answer: c. American exercise can also change timing of the obligation.

149. [**Multiple response**] Bond-option risk can reflect:

a. Interest rates. b. Credit spreads and rating changes. c. Liquidity and embedded bond options. d. Sovereign/macroeconomic conditions.

Answer and explanation

Answer: a, b, c and d. Bond volatility may be lower than FX/commodity volatility but specific credit risk can be large.

150. [**Single choice**] Why is an OTC currency option especially sensitive to counterparty risk?

a. It has no notional value. b. FX markets never move. c. It is always cleared by a CCP. d. The contract may require delivery/settlement of large currency notional amounts.

Answer and explanation

Answer: d. Large notional and 24-hour market movement increase monitoring requirements.

151. [**Multiple response**] Structured warrants add which risks beyond a simple share holding?

a. Issuer credit. b. Leverage and time decay. c. Volatility, liquidity and settlement. d. Underlying market/currency risk.

Answer and explanation

Answer: a, b, c and d. They do not give ordinary voting/dividend rights.

11.2 Structured-product risks

152. [**Multiple response**] Structured products can expose investors to:

a. Market and structure risk. b. Credit/counterparty and liquidity risk. c. Early termination/reinvestment risk. d. Volatility, correlation, FX, legal and transactional risk.

Answer and explanation

Answer: a, b, c and d. Combining components creates layered risk.

153. [**Single choice**] In a CPPI structure that reaches the floor after a sharp fall, the fund may:

a. Move entirely to the risk-free asset and miss later risky-asset recovery. b. Automatically buy the risky asset at the low. c. Guarantee unlimited upside. d. Eliminate all issuer risk.

Answer and explanation

Answer: a. This is the cash-lock/gap-to-floor effect.

154. [**Single choice**] A structured note with a swap between issuer and another institution may expose the investor to:

a. Voting-rights risk. b. Both issuer and swap-counterparty credit, depending on the documents. c. No credit risk. d. Only the investor's own credit.

Answer and explanation

Answer: b. Determine who bears the swap counterparty risk.

155. [**Multiple response**] ISDA credit-event categories in the source include:

a. Bankruptcy. b. Failure to pay. c. Restructuring. d. Obligation acceleration/default and repudiation/moratorium.

Answer and explanation

Answer: a, b, c and d. The exact applicable definition is contractual.

156. [**Single choice**] Early termination risk means:

a. The investor receives an extra coupon for exiting. b. Only the underlying equity is affected. c. Selling before maturity may incur bond discount and derivative unwinding/break costs. d. The product always pays full principal immediately.

Answer and explanation

Answer: c. Principal preservation commonly applies only at maturity.

157. [**Single choice**] Reinvestment risk is greatest when:

a. A zero-coupon bond is held to maturity with no interim cash flows. b. Rates rise and all coupons are locked at a higher rate. c. There is no callable feature. d. Coupons/principal are returned and rates have fallen.

Answer and explanation

Answer: d. The returned cash may earn less than the original YTM.

158. [**Scenario**] A worst-of credit basket contains highly correlated banks. One bank defaults during a systemic shock. The main extra risk is:

a. Correlation/default clustering. b. Guaranteed diversification. c. Zero liquidity risk. d. Only a dividend adjustment.

Answer and explanation

Answer: a. High correlation can make several names deteriorate together.

159. [**Single choice**] Legal risk is relevant because:

a. A PHS replaces all legal documents. b. A note holder may own a claim on the issuer, not the underlying assets, and creditor priority matters. c. Every note holder automatically owns the collateral. d. Contract wording never affects outcome.

Answer and explanation

Answer: b. Recourse and priority are document-dependent.

11.3 Controls and 11.4 summary

160. [**Multiple response**] Futures risk limits may include:

a. Open contracts. b. Maximum loss. c. Maturity and stress-test limits. d. No limits because futures are exchange traded.

Answer and explanation

Answer: a, b and c. Exchange liquidity does not remove firm-level exposure.

161. [**Calculation**] A bond has modified duration 2. Yield rises 100 bp. Approximate percentage price change is:

a. 0%. b. +2%. c. −2%. d. −200%.

Answer and explanation

Answer: c. $\Delta P/P = -D \times \Delta Y = -2 \times 1\% = -2\%$.

162. [**Calculation**] Bond price is $980 and PVBP is $1. Yield rises 5 bp. New approximate price is:

a. $980. b. $985. c. $930. d. $975.

Answer and explanation

Answer: d. Price falls $5 \times \$1 = \$5$.

163. [**Single choice**] A 1-day 95% VaR of SGD1m means:

a. There is model-estimated 5% probability of losing more than SGD1m over one day. b. The maximum possible loss is SGD1m. c. The portfolio earns SGD1m every day. d. The probability of any loss is 95%.

Answer and explanation

Answer: a. VaR is a threshold estimate, not a guarantee or a complete stress test.

164. [**Multiple response**] Option risk controls may set limits for:

a. Delta and gamma. b. Vega and theta. c. Rho. d. Maturity concentration, book size, premium and intraday exposure.

Answer and explanation

Answer: a, b, c and d. Greek limits should be linked to scenarios and hedging capacity.

165. [**Single choice**] A CSA primarily helps manage:

a. A fund's dividend. b. Collateral terms for OTC counterparty exposure. c. An ordinary stock split. d. Exchange index construction.

Answer and explanation

Answer: b. Settlement/confirmation controls remain necessary.

Chapter 12 — Case Study Applications

12.1 DCI and Chapter 7 application

166. [**Calculation**] A USD1m DCI pays 11% p.a. for three months. If no conversion applies, principal plus interest is:

a. USD1,011,000. b. USD1,027,500. c. USD1,110,000. d. USD1,330,000.

Answer and explanation

Answer: b. Interest = $1{,}000{,}000 \times 11\% \times 3/12 = 27{,}500$.

167. [**Scenario**] A client cannot accept receiving another currency. The best response is:

a. Hide the conversion clause. b. Treat conversion as a free benefit. c. Explain that a DCI may be unsuitable because conversion is part of the payoff. d. Recommend it because the headline yield is high.

Answer and explanation

Answer: c. Suitability depends on the client's willingness and ability to own the alternate currency.

12.2 Brian and Chapter 9/11 application

168. [**Multiple response**] Before Brian trades a leveraged downside/volatility strategy, check:

a. Risk willingness and understanding. b. External positions, capacity and credit limits. c. Buffers, margin and stop/offset discipline. d. Current research/market information and profit-taking plan.

Answer and explanation

Answer: a, b, c and d. Aggressive risk appetite does not justify maximising leverage.

169. [**Single choice**] If Brian expects a sharp fall and volatility rise, which is generally directionally consistent?

a. Short a volatility ETF. b. Buy a short-term bullish call only. c. Buy a range-bound short strangle. d. Buy an index put and/or long volatility ETF, subject to risks.

Answer and explanation

Answer: d. Timing, theta, contango and premium risk must still be explained.

12.3 William and Chapter 5/3 application

170. [**Multiple response**] William's portfolio-protection choices can include:

a. Buying tailored protective puts. b. Shorting a correlated MSCI Singapore future. c. Lending shares for a fee. d. Buying an ELN to protect against a fall.

Answer and explanation

Answer: a, b and c. A long ELN/short-put-like structure generally adds downside conversion exposure rather than protection.

171. [**Single choice**] Why is short MSCI Singapore futures an imperfect hedge for William?

a. His three stocks form only part of the index and their correlation is not one. b. Futures cannot be shorted. c. Futures have no MTM. d. The portfolio has no equity exposure.

Answer and explanation

Answer: a. Basis and composition risk remain.

12.4 Mr Sunjoyo and Chapter 8 application

172. [**Calculation**] An ELN costs SGD975,000 and converts into 80,000 shares. The break-even share price is:

a. SGD10.00. b. SGD12.19. c. SGD12.50. d. SGD13.30.

Answer and explanation

Answer: b. `975,000/80,000 = 12.1875`, approximately SGD12.19.

173. [**Scenario**] If the shares finish at SGD12.00, below the strike, the investor should understand that:

a. The note becomes a deposit-insured product. b. The headline 10.66% yield guarantees profit. c. The share value is SGD960,000, below the SGD975,000 purchase price before other costs. d. The investor automatically receives SGD1m cash.

Answer and explanation

Answer: c. A short-put-like ELN can lose principal.

12.5 Robert and Chapter 8/7 application

174. [**Calculation**] A range accrual pays 6% inside for two months and 1% outside for one month of a three-month period. Weighted annualised result is closest to:

a. 6.00%. b. 1.00%. c. 2.67%. d. 4.33%.

Answer and explanation

Answer: d. `6×2/3 + 1×1/3 = 4.33%`.

175. [**Multiple response**] To make Robert's digital spread cheaper or raise its maximum rate, one might:

a. Narrow the range. b. Skew the range toward one side. c. Lower the minimum yield. d. Increase maturity, subject to the product economics.

Answer and explanation

Answer: a, b, c and d. Widening the range generally makes the inside condition more likely and more expensive.

12.6 Harry and Chapter 7 application

176. [**Calculation**] An accumulator delivers 10,000 shares each day and knocks out on day 13, excluding the knock-out day. Shares accumulated are:

a. 10,000. b. 120,000. c. 130,000. d. 250,000.

Answer and explanation

Answer: b. Days 1–12 × 10,000 = 120,000.

177. [**Multiple response**] Accumulator risks include:

a. Purchase obligation below market price. b. Capped upside if knock-out occurs. c. Margin call/break cost. d. Larger quantity under 1×2 gearing.

Answer and explanation

Answer: a, b, c and d. The investor should be willing to own the maximum quantity at the strike.

12.7 Brandon and Chapter 4 application

178. [**Single choice**] Brandon expects a large gold move but is uncertain of direction. The most direct strategy is:

a. Long straddle. b. Short straddle. c. Short put only. d. Covered call only.

Answer and explanation

Answer: a. Long call plus long put benefits from a sufficiently large move after premiums.

179. [**Multiple response**] Two-sided volatility combinations can include:

a. Bull call spread plus bear put spread. b. Bull call spread plus bear call spread. c. Two bearish spreads only. d. Two bullish spreads only.

Answer and explanation

Answer: a and b. The objective is to have one bullish and one bearish component, subject to cost/overlap.

12.8 Tracy and Chapter 2 application

180. [**Single choice**] A CMS differs from a vanilla IRS because the floating portion resets against:

a. A fixed bond price with no reset. b. A constant-maturity long-term rate. c. A company's dividend. d. The spot equity price only.

Answer and explanation

Answer: b. CMS expresses views on the yield-curve shape.

181. [**Scenario**] Tracy expects short rates to rise faster than long rates and the curve to flatten. The source's strategy is:

a. Buy a gold straddle. b. Sell all fixed income. c. Pay the long-end rate and receive the short-end rate. d. Pay the short-end rate and receive the long-end rate.

Answer and explanation

Answer: c. The trade targets the expected relative movement of the two ends, subject to swap conventions.

Rapid mixed review

182. [**Multiple response**] Which products can create a large exposure from a relatively small initial outlay?

a. Futures. b. CFDs. c. Warrants/DLCs. d. Some accumulators and structured notes.

Answer and explanation

Answer: a, b, c and d. Leverage may magnify both return and loss.

183. [**Single choice**] The first question to ask about any derivative payoff is:

a. Is the coupon headline high? b. Is it listed in a brochure? c. What is the marketing name? d. What is the underlying and what position/exposure does the investor have?

Answer and explanation

Answer: d. Underlying, direction and trigger determine the risk map.

184. [**Multiple response**] Which are examples of "protection sold" by an investor seeking yield?

a. Selling a put in a bull ELN. b. Selling a call in a yield-enhancement note. c. Selling credit protection in a CLN. d. Selling a receiver/payer swaption in a structured product.

Answer and explanation

Answer: a, b, c and d. The enhanced yield is compensation for contingent liability.

185. [**Single choice**] Which statement is safest for exam calculations?

a. Identify the day-count, notional, multiplier, tick, conversion ratio and settlement convention first. b. Use 365 days for every product. c. Round before calculating the payoff. d. Ignore the contract multiplier.

Answer and explanation

Answer: a. The source uses 365 days for Singapore Government Securities and 360 days for other products unless stated otherwise.

186. [**Multiple response**] A complete suitability explanation should cover:

a. Best/base/worst-case scenarios. b. Liquidity and early-exit rules. c. Issuer/counterparty and collateral/recourse. d. Fees, leverage, margin and currency.

Answer and explanation

Answer: a, b, c and d. Good advice explains how the client can lose, not only how the product can earn.

Chapter 5 — Structured Warrants and Daily Leveraged Certificates (expanded)

187. [**Single choice**] Company warrants differ from structured warrants in that company warrants are:

a. Always longer than six months. b. Call only, while structured warrants may be calls or puts. c. Settled only in cash. d. Issued by third parties with no dilution.

Answer and explanation

Answer: b. Structured warrants may be calls or puts and are usually under six months; company warrants are call only.

188. [**Calculation**] A warrant has share price SGD 18.16, warrant price SGD 0.325 and conversion ratio 8:1. Gearing is:

a. 55.9×. b. 5.58×. c. 6.98×. d. 8.00×.

Answer and explanation

Answer: c. `18.16 / (0.325 × 8) = 6.98×`.

189. [**Single choice**] DBS shares rise 0.22% while the DBS call warrant falls 6.15% over the same period. The most likely explanation is:

a. Gearing was below 1×. b. The conversion ratio was removed. c. The issuer defaulted. d. Time decay outweighed the small favourable move.

Answer and explanation

Answer: d. A short-dated warrant loses time value, which can exceed the gain from a small favourable underlying move.

190. [**Calculation**] A new buyer pays SGD 0.665 for a call warrant with strike SGD 16.50 and conversion ratio 8:1. The breakeven share price is:

a. SGD 21.82. b. SGD 22.19. c. SGD 16.50. d. SGD 17.17.

Answer and explanation

Answer: a. `0.665 × 8 + 16.50 = 21.82`.

191. [**Single choice**] For SGX structured warrants, exercise style and settlement are normally:

a. American style, physical delivery. b. European style, cash settled. c. Bermudan style, physical delivery. d. American style, cash settled.

Answer and explanation

Answer: b. SGX-listed structured warrants are European style and cash settled.

192. [**Multiple response**] Which statements about warrant price factors are correct?

a. A higher call exercise price makes the call worth less. b. A higher put exercise price makes the put worth more. c. Higher interest rates raise call warrant prices. d. Higher interest rates raise put warrant prices.

Answer and explanation

Answer: a, b and c. Higher rates raise call prices because hedging costs rise, and lower put prices because put hedging becomes cheaper.

193. [**Single choice**] An issuer that commits to make a market in its structured warrants is exempt from:

a. The requirement to appoint a warrant agent. b. All disclosure obligations. c. Minimum placement and holder-size requirements, with a SGD 2 million minimum issue size. d. Issuer credit risk.

Answer and explanation

Answer: c. The DMM commitment substitutes for placement and holder-spread requirements, and the minimum issue size is SGD 2 million, reduced from SGD 5 million.

194. [**Single choice**] SGX Asian-style warrant expiry means settlement is based on:

a. The highest price during the warrant's life. b. The exercise price. c. The single closing price on the expiry date. d. The average of official closing prices over the five market days before expiry.

Answer and explanation

Answer: d. The last trading day is at least three business days before expiry, and the settlement price is a five-day average.

195. [**Calculation**] A 5× long DLC's underlying falls 10% and the DLC falls 50%. After the 30-minute airbag suspension the underlying falls a further 5%. The DLC price, from SGD 1.00 at the trigger, is approximately:

a. SGD 0.75. b. SGD 0.95. c. Zero. d. SGD 0.50.

Answer and explanation

Answer: a. A further 5% underlying move is 25% of SGD 1.00, so the DLC resets to SGD 0.75. Without the airbag it would be SGD 0.50.

196. [**Calculation**] An index rises 2% a day for five days. The compounded index gain is 10.41%. A 3× long DLC gains approximately:

a. 20.00%. b. 33.82%. c. 61.05%. d. 10.41%.

Answer and explanation

Answer: b. Daily resets compound: 3× produces 33.82%, while a 5× long produces 61.05%.

197. [**Multiple response**] Which are DLC risks?

a. Daily reset makes results path dependent. b. The airbag suspends trading and resets the reference level. c. A DMM may stand down in exceptional volatility. d. Capital is fully protected against a falling underlying.

Answer and explanation

Answer: a, b and c. A DLC can lose the entire invested capital; the maximum loss equals the amount invested but is not protected.

Chapter 6 — Barrier Options, Binary Options and CBBCs (expanded)

198. [**Single choice**] Breaching a knock-out barrier is more damaging than a capped spread because:

a. The knock-out extinguishes the whole option and loses the full intrinsic value accumulated. b. The knock-out converts the option into a future. c. A spread cannot be closed out. d. The spread has no maximum loss.

Answer and explanation

Answer: a. A spread merely caps the excess; a knock-out destroys the entire payoff.

199. [**Multiple response**] Which statements about knock-in barriers are correct?

a. The option is activated when the barrier is touched. b. Terms remain active from the touch through expiry even if price falls back. c. A rebate is usually paid if the barrier is never reached. d. The option is extinguished on a touch.

Answer and explanation

Answer: a, b and c. Extinguishing on a touch describes a knock-out, not a knock-in.

200. [**Single choice**] A barrier capital preservation certificate preserves:

a. Only the coupon. b. The purchase price at all times. c. The nominal amount at maturity only. d. The nominal amount throughout the product's life.

Answer and explanation

Answer: c. Capital preservation refers to nominal at maturity, not to the purchase price, and value may fall below it during the life.

201. [**Calculation**] A Category-R bull CBBC has underlying 1,060, strike 1,000, call price 1,040 and conversion ratio 10. What residual value does the holder receive when a mandatory call event occurs?

a. 0. b. 4. c. 6. d. 10.

Answer and explanation

Answer: b. Residual `= (call price − strike) / ratio = (1,040 − 1,000) / 10 = 4`. A Category-N contract sets its call price equal to the strike and so leaves nothing; only Category-R leaves a residual.

202. [**Multiple response**] Which statements about CBBC categories are correct?

a. A Category N CBBC has no residual value after an MCE. b. A Category R CBBC may pay a residual value, which may be zero. c. An MCE terminates the product before maturity. d. Category R guarantees a positive residual.

Answer and explanation

Answer: a, b and c. Category R does not guarantee a positive residual; it may be zero.

203. [**Scenario**] An R-bull CBBC bought at SGD 0.30 triggers an MCE and settles at a residual of SGD 0.05. The investor's loss is:

a. Zero, because residual value is paid. b. About 17%. c. About 50%. d. About 83%.

Answer and explanation

Answer: d. `(0.05 − 0.30) / 0.30 = −83.33%`, even though the underlying fell only about 12.5%.

Chapter 7 — Structured Deposits (expanded)

204. [**Multiple response**] Which are genuine client-suitability criteria for structured products?

a. Risk appetite, meaning both ability and willingness to bear risk. b. Investment experience and knowledge. c. Liquidity needs and investment horizon. d. The headline yield alone.

Answer and explanation

Answer: a, b and c. The headline yield is not a suitability criterion; the worst case must be understood.

205. [**Calculation**] A structured note has a call premium of USD 24 and the discount sum available from the zero-coupon bond is USD 21.65. The participation rate is:

a. 90.21%. b. 100.00%. c. 110.85%. d. 119.58%.

Answer and explanation

Answer: a. `21.65 / 24 = 0.9021 = 90.21%`. A bond with a 28.70 discount sum would give 119.58%.

206. [**Single choice**] In a standard dual currency investment, the investor is economically:

a. Long a call on the reference currency and short a deposit. b. Short a call on the reference currency and long a money market deposit. c. Long a put and long a bond. d. Holding a physical currency basket.

Answer and explanation

Answer: b. The premium on the sold option funds the yield above the money-market rate.

207. [**Calculation**] A range accrual pays 7.25% for 88 of 90 days inside the range and 2.00% outside. The payout is approximately:

a. 2.00%. b. 5.15%. c. 7.13%. d. 7.25%.

Answer and explanation

Answer: c. `(88/90 × 7.25%) + (2/90 × 2.00%) = 7.13%`.

208. [**Single choice**] In a 1×2 accumulator, once the price falls below the strike the investor must buy:

a. Nothing until maturity. b. Half the agreed quantity. c. The same quantity as above the strike. d. Twice the agreed quantity, subject to the terms.

Answer and explanation

Answer: d. The gearing magnifies the downside quantity, which is why the discounted strike is compensation for an obligation.

Chapter 8 — Structured Notes (expanded)

209. [**Calculation**] A range accrual note at 5% inside a SOR range accrues on 110 of 182 days. The half-year payout is approximately:

a. 3.02%. b. 5.00%. c. 5.50%. d. 1.50%.

Answer and explanation

Answer: a. `5% × 110/182 = 3.02%`.

210. [**Calculation**] An Apple ELN has strike USD 174.52 and an investment of USD 1,000,000. If conversion occurs, the investor receives approximately:

a. 1,745 shares. b. 5,730 shares. c. 5,000 shares. d. 6,000 shares.

Answer and explanation

Answer: b. `1,000,000 / 174.52 = 5,730 shares`.

211. [**Single choice**] A worst-of ELN's key additional risk is:

a. That the coupon is guaranteed. b. That principal is always protected. c. Correlation, because the weakest name drives the payoff. d. That the strongest name drives the payoff.

Answer and explanation

Answer: c. The payoff depends on the weakest constituent, so correlated names deteriorating together is the central risk.

212. [**Calculation**] A worst-of ELN's laggard trades at 29.16 against a conversion price of 27.00. The variable coupon is:

a. 9.09%. b. 27.00%. c. 2.80%. d. 8.00%.

Answer and explanation

Answer: d. `(29.16 / 27.00 − 1) × 100 = 8.00%`.

213. [**Single choice**] A credit-linked note investor is economically:

a. Selling credit protection on a reference entity. b. Holding the reference entity's equity. c. Holding a government bond. d. Buying credit protection.

Answer and explanation

Answer: a. The enhanced coupon compensates the investor for selling protection.

214. [**Multiple response**] Which are differences between a bond-linked note and a credit-linked note?

a. A BLN is triggered by a bond price fall rather than a credit event. b. A BLN delivers the bond rather than referencing an ISDA credit event. c. A CLN references defined credit events. d. A BLN has no issuer risk.

Answer and explanation

Answer: a, b and c. A BLN still carries issuer risk.

215. [**Single choice**] An ETN differs from an ETF chiefly because an ETN is:

a. A segregated pool of assets. b. Unsecured listed debt with issuer credit risk. c. Deposit-insured. d. Always physically backed.

Answer and explanation

Answer: b. The investor has a claim on the issuer, not on a segregated pool.

Chapter 9 — Structured Funds (expanded)

216. [**Calculation**] In a CPPI fund, the portfolio is worth \$100 and the floor is \$85, with a crash size of 20%. The risky allocation is:

a. \$15. b. \$20. c. \$75. d. \$85.

Answer and explanation

Answer: c. Multiplier $= 1/0.20 = 5$; cushion $=$ \$15; risky $= 5 \times 15 =$ \$75.

217. [**Scenario**] A CPPI fund reaches its floor after a sharp fall. The most likely consequence is:

a. It removes issuer risk. b. It buys risky assets at the low. c. It moves entirely to the risk-free asset and may miss a recovery. d. It guarantees unlimited upside.

Answer and explanation

Answer: c. This is cash lock: the mechanism de-risks at the bottom.

218. [**Single choice**] A trustee must notify MAS of a breach within:

a. Two months. b. Three months. c. One business day. d. Three business days.

Answer and explanation

Answer: d. Three business days. The semi-annual report is due within two months and the annual report within three.

219. [**Multiple response**] Which are causes of an ETF's tracking error?

a. Transaction costs and cash drag. b. The replication method used. c. Futures roll in contango or backwardation. d. Hedging costs.

Answer and explanation

Answer: a, b, c and d. All four, plus portfolio holdings differing from the index.

220. [**Single choice**] Synthetic ETFs are subject to a net counterparty exposure limit of:

a. 10% of NAV, with collateral of at least 90% of NAV. b. 50% of NAV. c. There is no limit. d. 5% of NAV.

Answer and explanation

Answer: a. The 10% limit applies with at least 90% collateral, rebalanced daily.

Chapter 10 — CFDs (expanded)

221. [**Calculation**] An investor buys 10,000 CFDs at $2.00 and sells at $2.20, with 0.4% commission, 8% GST on commission and 5% p.a. financing for 10 days. Net profit is approximately:

a. $2,209.22. b. $1,790.78. c. $1,818.56. d. $2,000.00.

Answer and explanation

Answer: b. Gross $2,000 less $86.40 buy cost, $95.04 sell cost and $27.78 financing = $1,790.78.

222. [**Calculation**] A short CFD sells 10,000 at $3.30 and covers at $3.45, with $319.10 total expense and $6,600 margin. The ROI is approximately:

a. +4.55%. b. −4.55%. c. −27.56%. d. −100%.

Answer and explanation

Answer: c. A net loss of $1,819.10 on $6,600 margin is −27.56%, even though the underlying moved only 4.5%.

223. [**Single choice**] The predominant CFD business model in Singapore is:

a. A market maker quoting its own price. b. An exchange-traded CFD. c. A physical delivery model. d. Direct market access.

Answer and explanation

Answer: d. DMA routes orders through the provider into the underlying market.

224. [**Scenario**] A CFD provider applies a rollover adjustment to a position held overnight, and the client also holds a short position in a dividend-paying stock over the ex-date. The client should expect:

a. The provider to absorb both adjustments. b. No financing charge and a dividend credit. c. A rollover charge reflecting the financing differential, and a dividend debit because a short position pays the dividend. d. A dividend credit, because all positions receive the dividend.

Answer and explanation

Answer: c. A rolled CFD incurs the financing differential, and a short equity position is debited the dividend because it cannot deliver the actual entitlement.

Chapter 11 — Key Product and Investment Risks (expanded)

225. [**Multiple response**] Which are causes of basis risk in a futures hedge?

a. The hedged asset differs from the contract's underlying. b. The hedger is uncertain about the exact transaction date. c. The hedge is closed out well before expiry. d. The contract size does not match the hedged position.

Answer and explanation

Answer: a, b, c and d. All four, with the last causing under- or over-hedging.

226. [**Single choice**] A ”lock limit” market means:

a. The price has reached its daily limit and no counterparty will trade, so the position cannot be offset. b. The exchange has closed permanently. c. Margin requirements are suspended. d. The contract has been cash settled.

Answer and explanation

Answer: a. Without an offsetting trade there is no way to close the position.

227. [**Single choice**] Reinvestment risk is greatest for:

a. Zero-coupon bonds. b. Callable bonds with large coupons and long maturities. c. Short-dated discount instruments. d. Floating-rate notes resetting daily.

Answer and explanation

Answer: b. Zero-coupon bonds have none; callable high-coupon long bonds have the most.

228. [**Calculation**] A bond has modified duration 2 and the yield rises 100bp. The approximate price change is:

a. 0%. b. +2%. c. −2%. d. −20%.

Answer and explanation

Answer: c. $\Delta P/P \approx -D \times \Delta YTM = -2 \times 1\% = -2\%$.

229. [**Single choice**] A 1-day 95% VaR of SGD 1m means:

a. The portfolio will lose SGD 1m one day in five. b. The portfolio cannot lose more than SGD 1m. c. The maximum possible loss is SGD 1m. d. There is about a 5% probability of losing more than SGD 1m in one day.

Answer and explanation

Answer: d. VaR is a threshold estimate at a confidence level and horizon, not a maximum loss.

230. [**Single choice**] A short (sold) pay-fixed swaption exposes the seller to:

a. Unlimited loss if rates rise. b. No loss. c. Loss limited to the premium received. d. A loss capped at the fixed rate.

Answer and explanation

Answer: a. A receive-fixed swaption's loss is capped at the fixed rate, but a pay-fixed one is unlimited.

231. [**Multiple response**] Which are legitimate futures risk limits?

a. Open-contract limits. b. Maximum-loss limits. c. Maturity and stress-test limits. d. No limits, because futures are exchange traded.

Answer and explanation

Answer: a, b and c. Exchange trading does not remove firm-level risk.

232. [**Single choice**] A confirmation should always state:

a. Option style, expiry date and time, premium payment date, settlement date, strike and both currency amounts. b. Only the counterparty name. c. Only the commission. d. Only the trade date.

Answer and explanation

Answer: a. These six fields prevent settlement disputes.

Chapter 1 — Overview of Derivatives

233. [**Single choice**] A derivatives contract discharges its obligation by a payment of the difference rather than by delivering the underlying. Which term describes it?

a. Cash-settled b. Physically settled c. Fully collateralised d. Freely assignable

Answer and explanation

Answer: a. Cash settlement pays the value of the position; physical settlement transfers the underlying itself.

234. [**Multiple response**] Which of the following are functions of a central clearing house?

a. Interposing itself as central counterparty to each trade b. Mutualising default risk through a default fund c. Guaranteeing that every cleared trade is profitable d. Collecting margin from its clearing members

Answer and explanation

Answer: a, b and d. A clearing house novates the trade, mutualises default risk and collects margin. It cannot guarantee profitability, because prices still move against positions after clearing.

235. [**Single choice**] Which statement best distinguishes an exchange-traded derivative from an OTC derivative?

a. Exchange-traded contracts are standardised and centrally cleared; OTC contracts are negotiated bilaterally b. OTC contracts are always clearer than exchange-traded contracts c. Exchange-traded contracts carry no risk of any kind d. Only OTC contracts require the posting of margin

Answer and explanation

Answer: a. Standardisation plus central clearing define the exchange-traded market. Exchange-traded contracts still carry clearing-house exposure, and OTC trades may be cleared voluntarily.

236. [**Single choice**] A company buys a put option over a commodity that it produces and holds as inventory. The purpose of the trade is best described as:

a. Speculating on implied volatility b. Hedging a long physical position c. Arbitraging the futures curve d. Creating a synthetic bond position

Answer and explanation

Answer: b. A producer holding the physical commodity is long the underlying, so a put protects the value of that inventory against a price fall.

237. [**Multiple response**] Which risks are present in an OTC derivative and absent or reduced in an exchange-traded equivalent?

a. Bilateral counterparty credit risk on the other party b. Documentation and legal-enforceability risk c. Price risk on the underlying asset d. Liquidity risk from the absence of a central order book

Answer and explanation

Answer: a, b and d. Price risk exists in both markets. The OTC market adds bilateral credit exposure and documentation risk, and often offers poorer liquidity.

238. [**Calculation**] An underlying is quoted at 100. A derivative on it has a delta of 0.6. Approximately how much does the derivative price change if the underlying moves to 101?

a. 0.006 b. 0.06 c. 0.60 d. 1.60

Answer and explanation

Answer: c. Delta is the change in the derivative price per one-unit change in the underlying, so

$$0.6 \times 1 = 0.60$$

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239. [**Single choice**] Which statement about a forward contract is correct?

a. It is marked to market daily through a clearing house b. It is a private agreement with settlement deferred to a future date c. It can always be closed out on an exchange d. It needs no credit assessment of the counterparty

Answer and explanation

Answer: b. A forward is bilateral and deferred. It is not cleared and not exchange-tradable, so the counterparty must be assessed for credit.

240. [**Single choice**] Which of the following is not a derivative?

a. A newly issued ordinary share bought in an initial public offering b. An exchange-traded futures contract c. A currency forward d. A listed structured warrant

Answer and explanation

Answer: a. A share is a direct claim on the issuer. A future, a forward and a warrant all derive their value from an underlying.

241. [**Multiple response**] Which statements about the counterparty to an exchange-traded derivative are correct?

a. The clearing house becomes the counterparty to both sides b. The original counterparties no longer face each other directly c. The clearing house is exposed to its clearing members d. The exchange guarantees the credit of every member to every other member

Answer and explanation

Answer: a, b and c. Novation substitutes the clearing house for the original counterparties, so each side faces the clearing house and the clearing house faces its members. It is the clearing house, not the exchange, that carries the credit exposure.

242. [**Calculation**] A derivative position has a notional of SGD 10 million and is quoted at 2% of notional. What is the market value of the position?

a. SGD 200 b. SGD 20,000 c. SGD 200,000 d. SGD 2,000,000

Answer and explanation

Answer: c.

$$10{,}000{,}000 \times 2\% = 200{,}000$$

. The notional is a reference amount, not the value of the position.

Chapter 2 — Futures, Forwards & Swaps

243. [**Calculation**] A share trades at SGD 10.00. The six-month risk-free rate is 2% per annum, continuously compounded, and the share pays no dividend. What is the six-month forward price?

a. SGD 9.90 b. SGD 10.10 c. SGD 10.20 d. SGD 10.40

Answer and explanation

Answer: b.

$$F = Se^{rT} = 10.00 \times e^{0.02 \times 0.5} \approx 10.10$$

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244. [**Calculation**] A share trades at SGD 20.00 and will pay a dividend of SGD 0.50 in three months. The risk-free rate is 3% per annum. What is the fair three-month forward price, to the nearest cent?

a. SGD 19.65 b. SGD 19.85 c. SGD 20.15 d. SGD 20.65

Answer and explanation

Answer: a.

$$F = (S - PV(D))e^{rT} = (20.00 - 0.50)e^{0.03 \times 0.25} \approx 19.65$$

. The dividend lowers the forward price, since the forward holder does not receive it.

245. [**Single choice**] The cost-of-carry model implies that the futures price exceeds the spot price when:

a. The convenience yield exceeds the financing cost b. The financing and storage cost exceeds the income from holding the asset c. The underlying is in permanent backwardation d. Storage costs are negative

Answer and explanation

Answer: b. Contango, an upward-sloping curve, arises when the carry costs of financing and storage exceed the dividend or convenience yield.

246. [**Multiple response**] Which factors cause the futures price of a physical commodity to depart from the pure cost-of-carry value?

a. Storage and insurance costs b. A convenience yield from holding the physical c. The credit rating of the clearing member d. The level of interest rates

Answer and explanation

Answer: a, b and d. Storage, convenience yield and financing all enter the carry relationship. A clearing member's credit rating affects its margin, not the futures price.

247. [**Single choice**] In a plain vanilla interest rate swap, the party described as the fixed-rate payer is:

a. Paying fixed and receiving floating b. Paying floating and receiving fixed c. Long a fixed-rate bond and long a floating-rate note d. Paying fixed with no reference to a notional principal

Answer and explanation

Answer: a. The fixed-rate payer pays the fixed leg and receives the floating leg. Economically this is short a fixed-rate bond and long a floating-rate note.

248. [**Calculation**] An interest rate swap has a notional of SGD 50 million. The fixed rate is 3.0% and the floating rate fixes at 2.6% for the year. What is the net payment made by the fixed-rate payer?

a. SGD 150,000 b. SGD 200,000 c. SGD 1,300,000 d. SGD 1,500,000

Answer and explanation

Answer: b.

$$50{,}000{,}000 \times (3.0\% - 2.6\%) = 200{,}000$$

. The fixed payer pays because fixed exceeds floating.

249. [**Multiple response**] Which statements about a cross-currency swap are correct?

a. Principal amounts are exchanged at inception and re-exchanged at maturity b. Interest is paid in each of the two currencies c. Only the interest differential is ever exchanged d. It can change the currency exposure of an existing borrowing

Answer and explanation

Answer: a, b and d. A currency swap exchanges principal at both ends and pays interest in both currencies, so it can transform a borrowing from one currency into another. Exchanging only the differential describes an interest rate swap.

250. [**Single choice**] A non-deliverable forward differs from a deliverable forward because an NDF:

a. Has no fixing date at all b. Cash-settles the difference against a published fixing, with no delivery of the restricted currency c. Is always traded on an exchange d. Carries no counterparty risk

Answer and explanation

Answer: b. An NDF settles the difference between the contracted rate and a published fixing, which avoids any need to deliver a non-convertible currency.

251. [**Calculation**] An NDF on USD 5 million is struck at 7.5000 for USD/SGD. The fixing prints at 7.4500. Which party settles, and how much?

a. The party buying USD at 7.5000 receives SGD 250,000 b. The party buying USD at 7.5000 pays SGD 250,000 c. The party buying USD at 7.5000 receives SGD 2,500,000 d. No settlement occurs because the difference is too small

Answer and explanation

Answer: b. Buying USD at 7.5000 when the fixing is 7.4500 means overpaying, so that party pays the difference:

$$5{,}000{,}000 \times (7.5000 - 7.4500) = 250{,}000$$

.

252. [**Single choice**] In a cross-currency swap, the exchange of principal at maturity means the parties:

a. Re-exchange the principal amounts at the original rate, eliminating the currency exposure on the principal b. Re-exchange at the prevailing spot rate, creating a currency gain or loss c. Do not re-exchange principal d. Exchange only the interest differential

Answer and explanation

Answer: a. Re-exchange at the contractual rate is what strips out currency risk on principal; re-exchanging at spot would leave the exposure in place.

253. [**Multiple response**] Which statements about a forward rate agreement are correct?

a. It fixes an interest rate for a future period b. It is cash-settled against a reference rate at the start of the period c. It requires delivery of a bond d. It is an OTC contract between two parties

Answer and explanation

Answer: a, b and d. An FRA settles the interest differential in cash at the start of the notional period and involves no bond delivery.

254. [**Calculation**] A share trades at SGD 8.00. The one-year rate is 3% and the share pays a continuous dividend yield of 1%. What is the one-year forward price using continuous compounding?

a. SGD 8.16 b. SGD 8.24 c. SGD 8.32 d. SGD 8.40

Answer and explanation

Answer: a.

$$F = Se^{(r-q)T} = 8.00 \times e^{0.02} \approx 8.16$$

.

255. [**Single choice**] Backwardation in a commodity futures curve most commonly reflects:

a. A convenience yield from holding the physical that exceeds the carry cost b. A permanent decline in demand c. Negative interest rates only d. A futures exchange rule

Answer and explanation

Answer: a. When holders value the physical more than the futures price implies, the curve slopes down. That premium is the convenience yield.

Chapter 3 — Futures Strategies

256. [**Calculation**] A portfolio is worth SGD 10 million and has a beta of 1.2. A futures contract is priced at 3,000 with a multiplier of SGD 50. How many contracts reduce the beta to zero?

a. 67 b. 80 c. 133 d. 160

Answer and explanation

Answer: b.

$$N = \frac{\beta V}{Fm} = \frac{1.2 \times 10{,}000{,}000}{3{,}000 \times 50} = 80$$

.

257. [**Calculation**] Using the same portfolio and contract, how many contracts reduce beta from 1.2 to 0.6?

a. 20 b. 40 c. 60 d. 80

Answer and explanation

Answer: b. Only half the exposure is removed:

$$N = \frac{(1.2 - 0.6) \times 10{,}000{,}000}{3{,}000 \times 50} = 40$$

.

258. [**Single choice**] A short hedge in the futures market is appropriate when the hedger:

a. Will buy the asset in future b. Holds the asset and will sell it in future c. Has no exposure to the asset d. Wants to increase exposure to the asset

Answer and explanation

Answer: b. A short futures position offsets the risk that the price of an asset already held falls before it is sold.

259. [**Multiple response**] Which of the following make a futures hedge imperfect?

a. The hedged asset differs from the futures underlying b. The hedging horizon does not match the contract maturity c. The contract size cannot be matched exactly to the exposure d. The futures price converges to the spot price at expiry

Answer and explanation

Answer: a, b and c. Basis risk arises from asset, maturity and size mismatch. Convergence at expiry is what makes the hedge work rather than a defect in it.

260. [**Single choice**] A trader buys one futures contract and simultaneously sells a further-out contract in the same commodity. This position is:

a. An outright long position b. A calendar spread c. An inter-commodity spread d. A covered call

Answer and explanation

Answer: b. A calendar, or intra-commodity, spread holds opposite positions in two delivery months of the same commodity.

261. [**Calculation**] The minimum variance hedge ratio is 0.8. The exposure is SGD 4 million and each contract is worth SGD 100,000. How many contracts should be sold?

a. 25 b. 32 c. 40 d. 80

Answer and explanation

Answer: b.

$$N = h \times \frac{V}{C} = 0.8 \times \frac{4{,}000{,}000}{100{,}000} = 32$$

.

262. [**Single choice**] A strip hedge differs from a stack hedge because a strip:

a. Uses only the nearest contract and rolls it forward b. Matches contract maturities to each exposure period c. Requires no margin to be posted d. Carries no basis risk

Answer and explanation

Answer: b. A strip lines contracts up with the timing of each exposure. A stack uses bulk short-dated contracts and relies on rolling, which introduces roll risk.

263. [**Multiple response**] Which are legitimate reasons for a corporate to use exchange-traded futures rather than a forward?

a. Greater liquidity and the ability to exit before maturity b. Reduced counterparty exposure through central clearing c. Anonymity of the counterparty on the exchange d. Freedom from any obligation to post margin

Answer and explanation

Answer: a, b and c. Futures offer liquidity, central clearing and anonymity. They do not remove the margin obligation, which is the very mechanism that secures them.

264. [**Single choice**] A hedge ratio of 1.0 with a contract whose underlying exactly matches the exposure means:

a. The number of contracts equals the exposure divided by the contract value b. No contracts are needed c. Only a fraction of the exposure is hedged d. The hedge is over-insured

Answer and explanation

Answer: a. A ratio of 1.0 means the notional of futures equals the notional of the exposure, so the count is exposure divided by contract value.

265. [**Calculation**] An exposure of SGD 6 million is hedged with futures priced at 1,500 with a multiplier of SGD 100. The hedge ratio is 1.0. How many contracts are required?

a. 40 b. 45 c. 60 d. 90

Answer and explanation

Answer: a. Contract value

$$= 1{,}500 \times 100 = 150{,}000$$

, so

$$6{,}000{,}000/150{,}000 = 40$$

.

266. [**Multiple response**] Which statements about a tailing-the-hedge adjustment are correct?

a. It adjusts the number of contracts for the daily settlement of futures b. It accounts for the interest earned or paid on variation margin c. It increases the number of contracts relative to the untailed hedge d. It is unnecessary when the hedge is held to expiry

Answer and explanation

Answer: a, b and c. Tailing adjusts the ratio by the discount factor because variation margin is settled daily; for a hedger receiving margin the count rises. It applies while the position is open, including to expiry.

267. [**Single choice**] An inter-commodity spread takes positions in:

a. Two different but related commodities in the same delivery month b. The same commodity in two delivery months c. The cash and futures markets simultaneously d. Two unrelated asset classes

Answer and explanation

Answer: a. An inter-commodity spread pairs related commodities, such as two energy grades; pairing delivery months of one commodity is a calendar spread.

Chapter 4 — Options

268. [**Calculation**] A call option has a strike of 50 and the underlying is at 56 at expiry. The premium paid was 4. What are the payoff and profit per share?

a. Payoff 6, profit 2 b. Payoff 6, profit 6 c. Payoff 2, profit 2 d. Payoff 4, profit 0

Answer and explanation

Answer: a. Payoff

$$= \max(56 - 50, 0) = 6$$

and profit

$$= 6 - 4 = 2$$

, since the premium is recovered from the payoff.

269. [**Calculation**] A put option has a strike of 80 and the underlying is at 72 at expiry. The premium paid was 5. What is the profit per share?

a. 8 b. 5 c. 3 d. 0

Answer and explanation

Answer: c. Payoff

$$= \max(80 - 72, 0) = 8$$

and profit

$$= 8 - 5 = 3$$

.

270. [**Calculation**] A one-year call at strike 100 costs 3 and a one-year put at strike 100 costs 2. The risk-free rate is 2%. By put-call parity, what is the spot price?

a. SGD 95.00 b. SGD 98.20 c. SGD 99.02 d. SGD 105.00

Answer and explanation

Answer: c.

$$C - P = S - Ke^{-rT}$$

, so

$$S = 3 - 2 + 100e^{-0.02} = 1 + 98.02 = 99.02$$

.

271. [**Multiple response**] Which statements about American and European options are correct?

a. An American option may be exercised at any time up to expiry b. A European option may be exercised only at expiry c. An American call on a non-dividend-paying share is worth the same as the equivalent European call d. An American option is always worth less than the equivalent European option

Answer and explanation

Answer: a, b and c. Early exercise of an American call on a non-dividend payer is never optimal, so the values coincide. An American option also carries every right a European option has, so it can never be worth less.

272. [**Single choice**] A trader is long a call and short a put with the same strike and expiry. The position replicates:

a. A long forward struck at the option strike b. A short forward struck at the option strike c. A long straddle d. A covered call

Answer and explanation

Answer: a. By put-call parity a long call plus a short put replicates a long forward at the strike; the two premiums differ by the financing on the strike.

273. [**Calculation**] A share trades at 40. A one-year call at strike 42 costs 2.00 and a one-year put at strike 42 costs 3.50. What risk-free rate does put-call parity imply?

a. A negative rate b. Approximately zero c. About 1.2% d. Impossible to infer

Answer and explanation

Answer: c.

$$C - P = S - Ke^{-rT}$$

gives

$$-1.50 = 40 - 42e^{-r}$$

, so

$$42e^{-r} = 41.50$$

,

$$e^{-r} = 0.9881$$

and

$$r \approx 1.2\%$$

.

274. [**Single choice**] Which option position has an unlimited potential loss?

a. Long call b. Long put c. Short call d. Long straddle

Answer and explanation

Answer: c. A short call is exposed to unlimited appreciation in the underlying. A long put can lose only the premium, because the underlying cannot fall below zero.

275. [**Multiple response**] Which statements about theta are correct?

a. It is usually negative for a long option position b. It measures sensitivity to the passage of time c. Its magnitude is largest for at-the-money options close to expiry d. It is unaffected by the option's moneyness

Answer and explanation

Answer: a, b and c. Time decay erodes a long option and accelerates at the money as expiry approaches. Theta is strongly dependent on moneyness, so the last statement is false.

276. [**Calculation**] A straddle is bought with a call and a put both struck at 100, costing 4 and 3. What are the two break-even underlying prices?

a. 93 and 107 b. 96 and 104 c. 100 and 107 d. 93 and 100

Answer and explanation

Answer: a. Total premium is 7, so the position breaks even at

$$100 - 7 = 93$$

and

$$100 + 7 = 107$$

.

277. [**Single choice**] An increase in the risk-free rate will, other things equal:

a. Raise the value of a call and lower the value of a put b. Lower the value of a call and raise the value of a put c. Raise both d. Leave both unchanged

Answer and explanation

Answer: a. The forward price rises with the rate, which helps the call and hurts the put; the discounting of the strike has the opposite sign and reinforces that direction.

278. [**Calculation**] A call has a delta of 0.4 and a gamma of 0.02. The underlying rises by 2. What is the approximate change in the call price?

a. 0.80 b. 0.84 c. 0.88 d. 1.20

Answer and explanation

Answer: b.

$$\Delta \cdot dS + \tfrac{1}{2}\Gamma \cdot dS^2 = 0.4 \times 2 + 0.5 \times 0.02 \times 4 = 0.80 + 0.04 = 0.84$$

.

279. [**Multiple response**] Which statements about a butterfly spread are correct?

a. It combines a bull spread and a bear spread at adjacent strikes b. It profits most when the underlying finishes near the middle strike c. It has limited loss and limited profit d. It has unlimited profit potential

Answer and explanation

Answer: a, b and c. A butterfly is bounded on both sides: maximum profit at the middle strike and maximum loss capped at the net premium paid. It has no unlimited upside.

280. [**Single choice**] The buyer of a put option is, in effect:

a. Insured against a fall in the underlying, having paid a premium for that insurance b. Obliged to sell the underlying at the strike c. Exposed to unlimited loss d. Guaranteed a profit if the underlying falls

Answer and explanation

Answer: a. The put is a right rather than an obligation, so the loss is capped at the premium. A profit requires the fall to exceed the premium paid.

Chapter 5 — Structured Warrants and Daily Leveraged Certificates

281. [**Calculation**] A structured warrant has a conversion ratio of 5 warrants per share, an exercise price of SGD 2.00 and a warrant price of SGD 0.30. What is the effective cost per share on exercise?

a. SGD 1.50 b. SGD 2.30 c. SGD 3.50 d. SGD 0.30

Answer and explanation

Answer: c. Five warrants cost

$$5 \times 0.30 = 1.50$$

, plus the exercise price of 2.00, giving SGD 3.50 per share.

282. [**Calculation**] A daily leveraged certificate offers 3x the daily return of an index. The index rises 10% on day one and falls 10% on day two. What is the certificate's two-day return, ignoring costs?

a. 0% b. About -1% c. About -9% d. About -3%

Answer and explanation

Answer: c. Day one multiplies the value by 1.30. Day two applies -30% to the new level, so

$$1.30 \times 0.70 = 0.91$$

, a return of about -9%. A constant daily multiple does not return to par.

283. [**Single choice**] The key risk of holding a daily leveraged product over several weeks is:

a. The issuer may call the product at any time b. Volatility decay from daily rebalancing, so the return can diverge materially from a simple multiple of the underlying return c. The product is always physically settled d. The investor loses entitlement to dividends

Answer and explanation

Answer: b. Because leverage resets daily, returns compound along the day-to-day path rather than the cumulative move, so choppy markets erode value even if the underlying ends flat.

284. [**Multiple response**] Which statements about structured warrants are correct?

a. They are issued by a third party, usually a bank b. They are listed and traded on an exchange c. The creditworthiness of the issuer is a relevant risk d. They carry no expiry date

Answer and explanation

Answer: a, b and c. A structured warrant is issuer-created, exchange-listed and has a fixed expiry, so issuer credit and expiry both matter. Absence of an expiry date describes a perpetual instrument.

285. [**Calculation**] A call warrant with a 1:1 ratio is priced at SGD 0.25, has an exercise price of SGD 1.80 and the share trades at SGD 2.00. What is the premium over intrinsic value, as a percentage of the share price?

a. 0% b. 2.5% c. 12.5% d. 25%

Answer and explanation

Answer: b. Intrinsic value is

$$2.00 - 1.80 = 0.20$$

, so the premium is 0.05, which is 2.5% of the SGD 2.00 share price.

286. [**Single choice**] Which best describes gearing on a warrant?

a. The share price divided by the effective warrant cost per share b. The warrant price divided by the exercise price c. The exercise price divided by the share price d. The warrant's delta

Answer and explanation

Answer: a. Gearing measures the underlying exposure obtained per dollar of warrant premium, so it scales the share price against the cost of one share's worth of warrants.

287. [**Calculation**] A warrant has a delta of 0.5. An investor wants the same exposure as 2,000 shares. How many warrants are needed?

a. 1,000 b. 2,000 c. 3,000 d. 4,000

Answer and explanation

Answer: d. Warrants needed

$$= \text{share exposure}/\delta = 2{,}000/0.5 = 4{,}000$$

.

288. [**Single choice**] A daily leveraged certificate is reset each day. An investor holds it for a month in a market that ends exactly where it started but is highly volatile. The most likely outcome is:

a. A loss caused by volatility decay b. A gain equal to the leverage multiple c. No change in value d. A gain equal to the volatility

Answer and explanation

Answer: a. Daily rebalancing compounds along the path. In a round-trip volatile market the product loses value even though the underlying is unchanged.

289. [**Multiple response**] Which statements about the issuer of a structured warrant are correct?

a. The issuer is typically a bank or a financial institution b. The investor is exposed to the issuer's creditworthiness c. The issuer is the listed company whose shares underlie the warrant d. The issuer may provide liquidity by quoting bid and offer prices

Answer and explanation

Answer: a, b and d. A structured warrant is issued by a third party such as a bank, not by the underlying company, so the issuer's credit and its liquidity provision both matter.

Chapter 6 — Barrier Options, Binary Options and Callable Contracts

290. [**Single choice**] A knock-in option differs from a knock-out option because a knock-in:

a. Begins with no payoff potential and becomes alive when the barrier is touched b. Is extinguished when the barrier is touched c. Has no barrier at all d. Can only be European

Answer and explanation

Answer: a. A knock-in activates when the barrier is touched; a knock-out is extinguished on the same event.

291. [**Multiple response**] Which statements about a Category-R CBBC are correct?

a. Its call price is set away from the strike b. It may leave a residual value after a mandatory call event c. It leaves no residual value after a mandatory call event d. It carries no leverage

Answer and explanation

Answer: a and b. Category-R sets the call price apart from the strike so a residual can remain; Category-N leaves nothing. Both categories are leveraged instruments.

292. [**Calculation**] A Category-N bull CBBC has an underlying at 1,050, a call price of 1,000 and a strike of 1,000. A mandatory call event occurs. What does the holder receive?

a. Nothing b. The difference between 1,050 and 1,000 c. The strike amount of 1,000 d. A replacement contract

Answer and explanation

Answer: a. In a Category-N contract the call price equals the strike, so no residual value remains once the mandatory call event occurs.

293. [**Single choice**] A binary option pays a fixed amount if the underlying finishes above the strike and nothing otherwise. Its payoff profile is:

a. Linear in the underlying b. A step function c. Identical to a vanilla call d. Unbounded

Answer and explanation

Answer: b. A binary pays a fixed sum or nothing, so the payoff is discontinuous at the strike rather than linear.

294. [**Multiple response**] Which are genuine disadvantages of barrier options for the buyer?

a. The option can be extinguished by a move in the underlying b. The premium is lower than an equivalent vanilla option c. Secondary-market liquidity is often poorer than for vanilla options d. Valuation and ongoing risk monitoring are more complex

Answer and explanation

Answer: a, c and d. A lower premium is the advantage that compensates the buyer, not a disadvantage. The real drawbacks are knock-out risk, poorer liquidity and greater complexity.

295. [**Single choice**] A holder of a Category-N bull CBBC is most exposed to which event?

a. A small rise in the underlying b. A fall in the underlying to the call price, triggering a mandatory call and extinguishing the residual c. A change of the issuer's auditor d. A share split by the underlying issuer

Answer and explanation

Answer: b. The contract is designed to be called away when the underlying reaches the call price, at which point a Category-N holder receives nothing.

296. [**Calculation**] A down-and-out put has a strike of 100, a barrier of 80 and the underlying trades at 95. The option knocks out when the underlying touches 80. If it touches 80 and then recovers to 90, the holder has:

a. No payoff, because the option was extinguished at the barrier b. A payoff of 10 c. A payoff of 5 d. A payoff of 20

Answer and explanation

Answer: a. Knock-out is a path-dependent event: once the barrier is touched the option ceases to exist, regardless of where the underlying finishes.

297. [**Single choice**] The issuer of a CBBC protects itself against unlimited loss by:

a. Imposing a mandatory call event that terminates the contract when the underlying reaches the call price b. Charging a higher premium than a warrant c. Requiring the investor to post margin d. Listing the contract on an exchange

Answer and explanation

Answer: a. The mandatory call is the issuer's protection: it ends the contract before the underlying can move far past the strike, which is why the leverage in a CBBC is bounded by the call feature.

298. [**Multiple response**] Which risks arise specifically from the barrier feature of a knock-out option?

a. The option can be extinguished before expiry b. The option may become worthless even though the underlying finishes favourably c. The option is worth more than an equivalent vanilla option d. Valuing and hedging it requires monitoring the barrier continuously

Answer and explanation

Answer: a, b and d. A knock-out is cheaper than the equivalent vanilla option, not more valuable, which is precisely why the buyer accepts the extinction risk.

Chapter 7 — Structured Deposits and Other Structured Products

299. [**Single choice**] A structured deposit differs from an ordinary time deposit because a structured deposit:

a. Has a return that depends on the performance of an underlying b. Is always fully protected by deposit insurance c. Has no maturity date d. Cannot be sold by a bank

Answer and explanation

Answer: a. The defining feature is a return linked to an underlying. Depending on the structure, the principal itself may be at risk.

300. [**Multiple response**] Which of the following are structured products?

a. Equity-linked notes b. Callable range accrual notes c. A posted-rate fixed deposit taken at the branch counter d. Accumulators

Answer and explanation

Answer: a, b and d. A structured product embeds a derivative to shape its return. A posted-rate fixed deposit embeds no derivative, so it is not a structured product.

301. [**Calculation**] An equity-linked note offers 100% principal protection plus 60% participation in any rise in an index. The index rises 25% over the term. What total return does the investor receive?

a. 15% b. 25% c. 60% d. 85%

Answer and explanation

Answer: a.

$$60\% \times 25\% = 15\%$$

. The protection returns the principal; the participation rate scales the upside.

302. [**Single choice**] A dual currency investment exposes the investor primarily to:

a. Interest rate risk only b. Currency risk, because redemption may occur in the weaker of two currencies c. Equity price risk only d. No risk, because the deposit sits with a bank

Answer and explanation

Answer: b. A DCI pays an enhanced yield in exchange for the investor accepting conversion into an alternative currency at a set rate, so currency exposure is the core risk.

303. [**Multiple response**] Which factors must a representative consider before recommending a structured product?

a. The client's risk appetite and investment horizon b. The client's liquidity needs c. The credit risk of the product issuer d. The visual design of the marketing brochure

Answer and explanation

Answer: a, b and c. Suitability turns on appetite, horizon, liquidity and issuer credit. Presentation has no bearing on suitability.

304. [**Single choice**] Which structured product typically offers capital protection but limits participation in the upside?

a. A capital-protected note with a participation rate below 100% b. A daily leveraged certificate c. A Category-N CBBC d. A short call position

Answer and explanation

Answer: a. Protecting principal consumes optionality, so the investor normally receives a participation rate below 100%, or a capped return.

305. [**Calculation**] A structured product protects 95% of principal and participates at 100% in upside above the initial level. The index falls 10%. What return does the investor receive?

a. -10% b. -5% c. 0% d. +10%

Answer and explanation

Answer: b. The floor binds at 95% of principal, so a 10% index fall is absorbed down to the floor, a return of -5%.

306. [**Single choice**] An accumulator structure obliges the investor to buy the underlying at intervals. Its principal risk is that:

a. The investor must keep buying as the price falls, magnifying losses b. The investor may miss upside if the price rises quickly c. The structure has no maturity d. The investor receives no dividends under any circumstances

Answer and explanation

Answer: a. Accumulators typically have a knock-out that caps the upside while the obligation to buy continues on the downside, so the payoff is asymmetrically against the investor.

307. [**Multiple response**] Which statements about a structured deposit are correct?

a. The return is linked to an underlying such as an index, a rate or a currency b. Principal may be at risk depending on the structure c. It is covered by deposit insurance in full in every case d. It may restrict early withdrawal

Answer and explanation

Answer: a, b and d. A structured deposit links its return to an underlying and may put principal at risk. Deposit insurance treatment depends on the structure, so it cannot be assumed to apply in full.

308. [**Single choice**] A range accrual deposit pays an enhanced coupon conditional on a reference rate staying within a band. The investor is effectively:

a. Selling a digital or range option on the reference rate to the issuer b. Buying a call option on the reference rate c. Taking no market risk d. Hedging interest rate exposure

Answer and explanation

Answer: a. The enhanced coupon is compensation for writing optionality: the investor receives less when the rate leaves the band, which is the payoff of a sold range or digital option.

309. [**Calculation**] A structured deposit offers 100% principal protection after three years plus 40% of any rise in an equity index. The index rises 50%. What is the investor's total return over the term?

a. 15% b. 20% c. 40% d. 50%

Answer and explanation

Answer: b.

$$40\% \times 50\% = 20\%$$

over the term, with the principal returned in full.

310. [**Multiple response**] Which are appropriate suitability considerations for an OTC structured product sold to a retail client?

a. Whether the client understands the payoff profile and the downside scenario b. Whether the client's liquidity needs are compatible with the tenor c. Whether the issuer's credit standing is acceptable d. Whether the product has a recent marketing award

Answer and explanation

Answer: a, b and c. Understanding, liquidity and issuer credit are the substantive suitability questions. A marketing award has no bearing on whether the product suits the client.

311. [**Single choice**] The principal-protected floor in a structured product is best understood as:

a. A purchased option or a zero-coupon bond whose cost is financed by giving up some upside b. A guarantee provided free of charge by the issuer c. A regulatory requirement for all structured products d. A form of deposit insurance

Answer and explanation

Answer: a. Protection must be funded: either the portfolio is built around a zero-coupon bond that grows to par, or a protective put is purchased. Either way the cost comes out of the upside.

Chapter 8 — Structured Notes

312. [**Single choice**] A reverse convertible is best described as:

a. A bond together with a short put on the underlying b. A bond together with a long call on the underlying c. A pure equity position d. A floating-rate note with no embedded option

Answer and explanation

Answer: a. The issuer pays an enhanced coupon in return for the investor writing a put: if the underlying falls, the investor receives shares instead of par.

313. [**Calculation**] An equity-linked note has a notional of USD 1,000,000 and a strike of USD 50, with a conversion ratio equal to notional divided by strike. The underlying falls to USD 40 and the note converts. How many shares does the investor receive, and what are they worth?

a. 20,000 shares worth USD 800,000 b. 25,000 shares worth USD 1,000,000 c. 20,000 shares worth USD 1,000,000 d. 25,000 shares worth USD 800,000

Answer and explanation

Answer: a. Conversion shares

$$= 1{,}000{,}000/50 = 20{,}000$$

, worth

$$20{,}000 \times 40 = 800{,}000$$

, a 20% loss of principal.

314. [**Multiple response**] Which statements about a range accrual note are correct?

a. Coupon accrues only on days the reference rate is inside a defined range b. It offers a higher potential coupon than a plain note of the same maturity and issuer c. Coupon accrues on every day regardless of the range d. The investor bears the risk that the range is not satisfied

Answer and explanation

Answer: a, b and d. The enhanced coupon compensates the investor for the risk that the range is breached, in which case no coupon accrues for the affected days.

315. [**Calculation**] A range accrual note pays 4% per annum on a 360-day basis and the reference rate is inside the range on 250 of 360 days. What is the annual coupon?

a. 2.78% b. 4.00% c. 5.76% d. 1.39%

Answer and explanation

Answer: a.

$$4\% \times \frac{250}{360} = 2.78\%$$

.

316. [**Single choice**] The coupon on an inverse floater rises when:

a. The reference rate rises b. The reference rate falls c. The issuer's credit rating falls d. The note is called by the issuer

Answer and explanation

Answer: b. The coupon moves inversely to the reference rate, so the investor gains when rates fall and loses when rates rise.

317. [**Calculation**] An inverse floater pays 8% minus the three-month rate, floored at zero. If the three-month rate is 5.5%, what coupon does the investor receive?

a. 0% b. 2.5% c. 5.5% d. 8.0%

Answer and explanation

Answer: b.

$$8\% - 5.5\% = 2.5\%$$

, comfortably above the zero floor.

318. [**Single choice**] Which risk dominates a structured note with a long maturity and no secondary market?

a. Liquidity risk b. Reinvestment risk c. Settlement risk d. Operational risk

Answer and explanation

Answer: a. An instrument that cannot readily be sold exposes its holder to liquidity risk, which the issuer prices into the coupon.

Chapter 9 — Structured Funds

319. [**Single choice**] An exchange-traded fund differs from a unit trust primarily because an ETF:

a. Trades intraday on an exchange at market-determined prices b. Is not a collective investment scheme c. Holds no underlying assets d. Cannot be sold short under any circumstances

Answer and explanation

Answer: a. Both are collective schemes. The ETF's distinguishing feature is intraday exchange trading with creation and redemption by authorised participants.

320. [**Multiple response**] Which statements about constant proportion portfolio insurance (CPPI) are correct?

a. It allocates between a risky and a safe asset b. The allocation to the risky asset rises as portfolio value rises above the floor c. It guarantees the floor under every market condition, including gap moves d. It can be defeated by a market gap that jumps below the floor

Answer and explanation

Answer: a, b and d. CPPI scales exposure to the cushion above a floor. It cannot guarantee the floor through a gap, because it can trade only at prices actually available.

321. [**Calculation**] A CPPI portfolio is worth 100 with a floor of 80 and a multiplier of 3. What is the allocation to the risky asset?

a. 20 b. 40 c. 60 d. 80

Answer and explanation

Answer: c. Cushion

$$= 100 - 80 = 20$$

, so the risky allocation is

$$3 \times 20 = 60$$

.

322. [**Single choice**] A closed-end fund differs from an open-end fund because a closed-end fund:

a. Issues a fixed number of shares that trade on an exchange b. Always redeems shares at net asset value on demand c. Has no net asset value at all d. May not use leverage under any circumstances

Answer and explanation

Answer: a. A closed-end fund has a fixed share count, so its market price can diverge from net asset value. An open-end fund issues and redeems at NAV.

323. [**Multiple response**] Which are limitations of using the Sharpe ratio to compare funds?

a. It assumes returns are normally distributed b. It penalises upside volatility as well as downside volatility c. It is entirely independent of the fund's fees d. It can be distorted by smoothing of reported returns

Answer and explanation

Answer: a, b and d. Sharpe treats all dispersion as risk, is sensitive to non-normality and sensitive to return smoothing. Fees reduce the numerator, so the ratio is not independent of them.

324. [**Single choice**] The tracking error of an ETF measures:

a. The volatility of the difference between fund return and index return b. The fund's total expense ratio c. The prevailing bid-ask spread d. The fund's level of leverage

Answer and explanation

Answer: a. Tracking error is the dispersion of the fund-minus-index return; a low figure indicates close replication of the index.

325. [**Single choice**] A fund's net asset value per share is computed as:

a. Total assets less total liabilities, divided by the number of shares outstanding b. Total assets divided by the number of shares, ignoring liabilities c. The market price multiplied by the number of shares d. The subscription price set at launch

Answer and explanation

Answer: a. NAV nets liabilities against assets before dividing by shares. The market price of a closed-end fund can differ from NAV; the launch price is a single historical figure.

326. [**Multiple response**] Which statements about ETFs and their creation and redemption mechanism are correct?

a. Authorised participants can create new ETF shares by delivering the underlying basket b. Creation and redemption help keep the market price close to NAV c. Only institutional investors may hold ETF shares d. Redemption involves exchanging ETF shares for the underlying assets

Answer and explanation

Answer: a, b and d. The creation and redemption arbitrage is what anchors the market price to NAV, and it operates through authorised participants. Retail investors buy ETF shares on exchange like any other listed security.

327. [**Single choice**] A structured fund with capital protection typically achieves that protection by:

a. Investing a large part of the portfolio in a safe asset and using the remainder to buy derivative exposure b. Guaranteeing the return by contract with the investor c. Holding only cash d. Borrowing to buy more of the risky asset

Answer and explanation

Answer: a. Protection is engineered by funding a floor with a safe asset and spending only the residual on optionality. Any guarantee is a credit exposure to the guarantor rather than a free lunch.

328. [**Calculation**] A CPPI structured fund has a value of 1,000, a floor of 800 and a multiplier of 2. The risky asset then falls 50% while the safe asset is flat. What is the new portfolio value?

a. 600 b. 800 c. 900 d. 1,000

Answer and explanation

Answer: b. The cushion was

$$1{,}000 - 800 = 200$$

, so the risky allocation was

$$2 \times 200 = 400$$

and the safe allocation 600. A 50% fall in the risky asset destroys 200, leaving

$$600 + 200 = 800$$

— exactly the floor. A larger gap would break through it, because CPPI can only trade at prices actually available.

329. [**Multiple response**] Which are genuine disadvantages of ETFs relative to an index unit trust?

a. Bid-ask spreads and brokerage are incurred on each purchase b. The market price can diverge from NAV in stressed markets c. They cannot be sold during the trading day d. Small purchases may suffer proportionally higher dealing costs

Answer and explanation

Answer: a, b and d. Trading costs and possible NAV divergence are the trade-offs for intraday liquidity. The ability to sell during the day is an advantage of ETFs, so the third statement is false.

330. [**Single choice**] A fund's returns are smoothed by the manager. Which performance measure is most distorted?

a. The Sharpe ratio, because measured volatility is understated b. The total expense ratio c. The fund's NAV d. The fund's tracking error against a cash benchmark

Answer and explanation

Answer: a. Smoothing suppresses measured volatility, which inflates the Sharpe ratio. The expense ratio and NAV are unaffected by smoothing.

Chapter 10 — Contracts for Differences (CFDs)

331. [**Calculation**] An investor buys 5,000 CFDs at SGD 4.00 with a 10% margin requirement. What initial margin is required?

a. SGD 200 b. SGD 2,000 c. SGD 20,000 d. SGD 200,000

Answer and explanation

Answer: b. Position value

$$= 5{,}000 \times 4.00 = 20{,}000$$

, so margin

$$= 10\% \times 20{,}000 = 2{,}000$$

.

332. [**Calculation**] The same 5,000 CFDs are closed at SGD 3.80. What is the realised profit or loss, ignoring financing and commission?

a. A loss of SGD 200 b. A loss of SGD 1,000 c. A gain of SGD 1,000 d. A gain of SGD 200

Answer and explanation

Answer: b.

$$5{,}000 \times (3.80 - 4.00) = -1{,}000$$

, a loss of SGD 1,000 on SGD 2,000 of margin — a 50% loss of equity for a 5% adverse move.

333. [**Single choice**] A CFD provider applies an overnight financing charge. The charge is best described as:

a. A one-off commission charged on opening the position b. A daily interest adjustment for the leveraged portion of the position c. A statutory levy collected for the regulator d. A dividend adjustment made to the account

Answer and explanation

Answer: b. Because a CFD is leveraged, the provider finances the balance and charges interest daily, usually at a reference rate plus a spread.

334. [**Multiple response**] Which are true differences between a CFD and an exchange-traded futures contract?

a. CFDs are usually traded OTC; futures are exchange-traded b. Futures have standardised terms; CFDs are bespoke c. A CFD carries no counterparty risk at all d. A CFD position can be held open indefinitely, subject to the provider's terms

Answer and explanation

Answer: a, b and d. A CFD is a bilateral contract with the provider, so counterparty risk is present rather than absent. Open-ended duration and bespoke terms are the genuine differences.

335. [**Calculation**] An investor is short 2,000 CFDs from SGD 5.00. A dividend of SGD 0.10 per share goes ex. What adjustment applies?

a. The investor receives SGD 200 b. The investor pays SGD 200 c. The investor receives SGD 20 d. The investor pays SGD 20

Answer and explanation

Answer: b. A short CFD holder compensates the provider for the dividend the provider must pay:

$$2{,}000 \times 0.10 = 200$$

paid by the investor.

336. [**Single choice**] Which is the most accurate statement about a guaranteed stop on a CFD?

a. It is provided free of charge on every position b. It guarantees the exit price but usually costs a wider spread or an explicit premium c. It removes all market risk from the position d. It is available only on exchange-traded futures

Answer and explanation

Answer: b. The provider absorbs gap risk on a guaranteed stop, so it charges for that protection through a wider spread or a fee.

337. [**Multiple response**] Which risks are specific to trading CFDs rather than holding the underlying shares outright?

a. Leverage can magnify losses beyond the initial outlay b. The provider may become insolvent c. The position may be closed out on a margin call d. The investor receives dividends directly from the company

Answer and explanation

Answer: a, b and c. A CFD holder has no direct shareholder entitlement; any dividend reaches the account only as a contractual adjustment from the provider.

338. [**Calculation**] A CFD account has equity of SGD 5,000 and a maintenance margin requirement of SGD 4,000. By how much can the position lose before a margin call is triggered?

a. SGD 1,000 b. SGD 4,000 c. SGD 5,000 d. SGD 9,000

Answer and explanation

Answer: a. A margin call follows once equity falls to the maintenance requirement:

$$5{,}000 - 4{,}000 = 1{,}000$$

of cushion.

339. [**Single choice**] A CFD provider's terms state that the client is the beneficial owner of the underlying. This is:

a. A misleading statement, because a CFD confers no beneficial ownership of the underlying b. Correct, because the CFD is collateralised c. Correct for long positions only d. Correct if the provider holds the shares

Answer and explanation

Answer: a. A CFD is a contract for the difference: it creates no ownership, no voting rights and no direct dividend entitlement. Any dividend reaches the account only as a contractual adjustment.

340. [**Multiple response**] Which statements about the financing cost of a CFD are correct?

a. It is charged on the leveraged portion of the position b. It is usually referenced to an interbank rate plus a spread c. A long position generally pays financing and a short position may receive it d. It is a one-off charge at inception

Answer and explanation

Answer: a, b and c. Financing accrues daily on the financed amount, referenced to a market rate plus the provider's spread. A short position may earn a rate reflecting the collateral posted.

341. [**Calculation**] A long CFD position of SGD 50,000 is financed at 4% per annum. What is the approximate daily financing charge, using a 360-day year?

a. SGD 1.67 b. SGD 5.56 c. SGD 55.60 d. SGD 200.00

Answer and explanation

Answer: b.

$$50{,}000 \times 4\%/360 = 5.56$$

per day.

342. [**Single choice**] Which is the most accurate description of a CFD provider's conflict of interest with its client?

a. The provider is typically the counterparty to the client's trade, so the client's loss can be the provider's gain b. The provider earns only commission and has no directional exposure c. The provider acts as an exchange and matches clients anonymously d. There is no conflict because prices come from the underlying market

Answer and explanation

Answer: a. Where the provider is the counterparty rather than an agent, the client's loss is the provider's gain, which is why hedging, disclosure and conduct rules matter.

Chapter 11 — Key Products and Investment Risks for Derivatives and Structured Products

343. [**Multiple response**] Which are key risks of investing in structured products?

a. Market risk on the underlying b. Liquidity risk on the note itself c. Issuer credit risk d. Currency risk where the note is denominated in a foreign currency

Answer and explanation

Answer: a, b, c and d. All four apply: the underlying can move against the investor, the note may be illiquid, the issuer may default, and a foreign-currency note adds currency exposure.

344. [**Single choice**] Which risk is most closely associated with holding a single large bespoke OTC derivative position?

a. Counterparty credit risk b. Deposit insurance risk c. Board composition risk d. Dividend timing risk

Answer and explanation

Answer: a. A large bespoke OTC position concentrates exposure on one counterparty, making that counterparty's creditworthiness the dominant incremental risk.

345. [**Multiple response**] Which statements about basis risk are correct?

a. It arises when the cash asset differs from the futures underlying b. It remains if the hedge is lifted before the contract expires c. It narrows towards zero as the contract approaches expiry d. It is eliminated by using a larger number of contracts

Answer and explanation

Answer: a, b and c. Basis converges at expiry, so an early unwind leaves residual basis risk. Adding contracts changes the size of the exposure, not the basis.

346. [**Single choice**] A credit default swap primarily transfers:

a. Interest rate risk b. Credit risk on a reference entity c. Equity price risk d. Operational risk

Answer and explanation

Answer: b. The protection buyer pays a premium and receives a payment on a defined credit event affecting the reference entity.

347. [**Multiple response**] Which are ISDA-recognised credit events for a credit default swap?

a. Failure to pay b. Restructuring c. Obligation acceleration d. A fall in the reference entity's share price

Answer and explanation

Answer: a, b and c. Standard credit events include failure to pay, restructuring and obligation acceleration. A share-price fall is not a credit event, although it may reflect deteriorating credit.

348. [**Single choice**] Reinvestment risk is most relevant to which investor?

a. One holding a long-dated zero-coupon bond to maturity b. One holding a short-dated coupon bond and intending to roll it over c. One holding physical gold in a vault d. One holding a perpetual floating-rate note

Answer and explanation

Answer: b. Reinvestment risk bites when maturing cash must be placed at unknown future rates, which affects a rolling coupon portfolio.

349. [**Multiple response**] Which risk-management actions are appropriate for a leveraged derivatives portfolio?

a. Setting and enforcing stop-loss levels b. Imposing a daily loss limit c. Measuring exposure with value at risk d. Removing all position limits so that no forced exit can occur

Answer and explanation

Answer: a, b and c. Stops, daily loss limits and VaR measurement are the standard controls. Removing limits converts a managed loss into an unmanaged one.

350. [**Single choice**] A position has a one-day 95% value at risk of SGD 50,000. What does this mean?

a. The maximum possible loss is SGD 50,000 b. Under the model's assumptions, there is a 5% chance of losing more than SGD 50,000 over one day c. The position will lose exactly SGD 50,000 on one day in twenty d. The expected loss over one day is SGD 50,000

Answer and explanation

Answer: b. VaR is a quantile, not a maximum: it states a loss threshold expected to be exceeded with the stated probability over the stated horizon, conditional on the model.

351. [**Single choice**] A counterparty to a long-dated OTC swap is downgraded sharply. Which risk has most directly increased?

a. Credit risk on the position b. Market risk on the underlying c. Operational risk d. Reinvestment risk

Answer and explanation

Answer: a. A downgrade raises the probability of default on the exposure, so the credit risk on the in-the-money position rises. The underlying has not moved.

352. [**Multiple response**] Which statements about currency risk in structured products are correct?

a. A note denominated in a foreign currency exposes the investor to the exchange rate on both coupon and principal b. Hedging the currency removes the exposure but may cost yield c. Currency risk is irrelevant if the issuer is highly rated d. A dual currency structure deliberately embeds currency exposure

Answer and explanation

Answer: a, b and d. Issuer creditworthiness has no bearing on exchange-rate movements, so a strong issuer does not remove currency risk.

353. [**Single choice**] Which risk is most relevant to an investor who may need to sell a structured note before maturity?

a. Liquidity risk, because there may be no secondary market or only at a wide discount b. Market risk only c. Operational risk d. Settlement risk

Answer and explanation

Answer: a. Early exit from a bespoke note depends on the issuer quoting a price, so the risk that the exit price is poor is a liquidity risk.

354. [**Calculation**] A portfolio of SGD 1,000,000 has a one-year 99% VaR of 8%. What loss does this indicate?

a. SGD 8,000 b. SGD 80,000 c. SGD 800,000 d. SGD 8,000,000

Answer and explanation

Answer: b.

$$1{,}000{,}000 \times 8\% = 80{,}000$$

, exceeded with a 1% probability over one year under the model.

Chapter 12 — Case Studies

355. [**Single choice**] A client with a two-year horizon and low risk tolerance wants exposure to equity upside. The most suitable structure is:

a. A daily leveraged equity certificate b. A capital-protected note with a participation rate c. A Category-N bull CBBC d. A short equity call position

Answer and explanation

Answer: b. A capital-protected note delivers upside while respecting a low risk tolerance and a defined horizon. Leveraged and CBBC structures can lose the whole investment.

356. [**Multiple response**] A client must settle a USD 2 million payable in six months. Which are appropriate hedging responses?

a. Buy a USD/SGD forward b. Buy a USD call option c. Buy a USD put option d. Enter a currency swap where there is a matching USD inflow

Answer and explanation

Answer: a, b and d. A USD payable is hedged by locking in the purchase of USD, whether by forward or by call. A USD put would add to the exposure, and a swap suits matched two-way flows.

357. [**Single choice**] A client's portfolio has drifted to 55% equity and 45% bonds against a 70/30 target. Which trade restores the target?

a. Sell equity and buy bonds b. Buy equity and sell bonds c. Sell both equity and bonds d. Take no action because the target is a long-term aim

Answer and explanation

Answer: b. The portfolio has drifted defensive relative to target, so equity must be bought and bonds sold to restore 70/30.

358. [**Calculation**] A client invests SGD 100,000 in a note paying a 4% fixed coupon plus 50% participation in any index return above a 10% hurdle. The index returns 20%. What total return does the client receive?

a. 4% b. 5% c. 9% d. 14%

Answer and explanation

Answer: c. Fixed coupon 4% plus

$$50\% \times (20\% - 10\%) = 5\%$$

, giving 9% in total.

359. [**Single choice**] A representative recommends a structured note without documenting the client's risk profile. The primary compliance failure is:

a. Failure to establish a reasonable basis for the recommendation b. Failure to hold a capital markets services licence c. Failure to publish a product highlights sheet d. Failure to notify MAS of the trade

Answer and explanation

Answer: a. Suitability requires a documented understanding of the client before recommending. Without it there is no reasonable basis for the recommendation.

360. [**Scenario**] A client holds a concentrated position in a single listed share and wants to protect it over three months without selling. Which is the most direct solution?

a. Buy a put option on the share, or sell a futures contract on it b. Buy a call option on the share c. Buy more of the share d. Invest in a money market fund

Answer and explanation

Answer: a. Protecting a long position requires a short exposure: a protective put or a short futures position. A long call adds to the exposure rather than hedging it.

361. [**Scenario**] A client wants a fixed, known cost for converting SGD to USD in six months but also wants to benefit if SGD strengthens. Which structure best fits?

a. A USD call option, which caps the cost while preserving the upside b. A forward, which locks the rate in both directions c. A USD put option d. A currency swap with no optionality

Answer and explanation

Answer: a. An option pays for certainty only if it is exercised, so it caps the cost while leaving the client free to transact at a better spot rate. A forward removes that freedom. A put would hedge the wrong direction.

362. [**Multiple response**] A retail client is offered a note with 120% participation and no principal protection. Which concerns are appropriate?

a. The client can lose part or all of the principal b. The enhanced participation is funded by the absence of protection c. The issuer's credit standing matters d. The note is suitable for any client because participation exceeds 100%

Answer and explanation

Answer: a, b and c. Participation above 100% is paid for by removing the floor, so the client bears full downside. Suitability must be assessed; a high participation rate is not itself an indicator of suitability.

363. [**Calculation**] A client has SGD 200,000 and wants at least SGD 195,000 returned after one year. If a one-year zero-coupon instrument yields 3%, how much is left to spend on optionality?

a. About SGD 5,000 b. About SGD 10,680 c. About SGD 15,534 d. About SGD 19,417

Answer and explanation

Answer: b. The sum needed today to reach the 195,000 floor in one year is

$$195{,}000/1.03 \approx 189{,}320$$

. The residual available for optionality is

$$200{,}000 - 189{,}320 \approx 10{,}680$$

.

Appendix D — Formulae sheet (consolidated)

364. [**Calculation**] Using the cost-of-carry model with continuous compounding, a share trades at 50, the risk-free rate is 4% and the dividend yield is 1%. What is the one-year futures price?

a. 48.51 b. 51.52 c. 52.00 d. 55.20

Answer and explanation

Answer: b.

$$F = Se^{(r-q)T} = 50 \times e^{0.03} \approx 51.52$$

.

365. [**Calculation**] An equity index futures contract has a multiplier of SGD 50 and a price of 3,200. How many contracts hedge a SGD 8 million portfolio with a beta of 0.9?

a. 45 b. 50 c. 60 d. 64

Answer and explanation

Answer: a.

$$N = \frac{0.9 \times 8{,}000{,}000}{3{,}200 \times 50} = 45$$

.

366. [**Calculation**] A bond portfolio has a modified duration of 6 and a value of SGD 20 million. The cheapest-to-deliver bond has a modified duration of 7.5 and the futures contract is priced at 110 with a multiplier of SGD 1,000. How many contracts are needed to hedge the portfolio?

a. 145 b. 150 c. 160 d. 175

Answer and explanation

Answer: a. The contract value is

$$110 \times 1{,}000 = 110{,}000$$

, so

$$N = \frac{6 \times 20{,}000{,}000}{7.5 \times 110{,}000} \approx 145.5$$

, that is 145 whole contracts.

367. [**Calculation**] A CPPI portfolio is worth 500 with a floor of 450 and a multiplier of 4. What is the allocation to the risky asset?

a. 50 b. 100 c. 200 d. 450

Answer and explanation

Answer: c. Cushion

$$= 500 - 450 = 50$$

, so the risky allocation is

$$4 \times 50 = 200$$

.

368. [**Calculation**] A range accrual note pays 6% per annum on a 30/360 basis and accrues on 75% of the days in a 180-day period. What coupon accrues for that period?

a. 2.25% b. 3.00% c. 4.50% d. 6.00%

Answer and explanation

Answer: a.

$$6\% \times \frac{180}{360} \times 75\% = 2.25\%$$

.

369. [**Calculation**] A CBBC has an underlying at 1,100, a strike of 1,000 and a call price of 1,050. What is the theoretical price of the contract?

a. 50 b. 100 c. 1,050 d. 1,100

Answer and explanation

Answer: a. The theoretical price of a CBBC is the underlying less the call price:

$$1{,}100 - 1{,}050 = 50$$

.

370. [**Calculation**] A warrant has a share price of 3.00, an exercise price of 2.40 and a conversion ratio of two warrants per share. What is the value of one warrant on exercise, ignoring premium?

a. 0.30 b. 0.60 c. 1.20 d. 2.40

Answer and explanation

Answer: a. Shares received per warrant are

$$1/2$$

, so value

$$= (3.00 - 2.40) \times \frac{1}{2} = 0.30$$

. Equivalently

$$\frac{3.00}{2} - \frac{2.40}{2}$$

.

Appendix A — CPPI strategy worked examples

371. [**Calculation**] A CPPI portfolio is worth 1,000 with a floor of 900 and a multiplier of 4. The risky asset then falls 10% while the safe asset is unchanged. What is the new portfolio value and the new risky allocation?

a. 960 and 240 b. 980 and 320 c. 1,000 and 400 d. 940 and 160

Answer and explanation

Answer: a. Risky was

$$4 \times 100 = 400$$

and safe was 600. Risky falls to 360, so the portfolio is

$$360 + 600 = 960$$

. The new cushion is

$$960 - 900 = 60$$

, giving a risky allocation of

$$4 \times 60 = 240$$

.

372. [**Calculation**] In the same CPPI example the risky asset instead rises 10%. What is the new portfolio value and the new risky allocation?

a. 1,040 and 440 b. 1,040 and 560 c. 1,100 and 400 d. 1,000 and 400

Answer and explanation

Answer: a. Risky rises to 440, so the portfolio is

$$440 + 600 = 1,040$$

. The cushion becomes

$$1{,}040 - 900 = 140$$

, giving a risky allocation of

$$4 \times 140 = 560$$

. The allocation of 560 refers to the next period; the value after the move is 1,040 with 440 currently in the risky asset.

Appendix B — MAS Product Highlights Sheet practice note

373. [**Single choice**] A product highlights sheet for a structured product must present:

a. The key features, risks and costs in a concise, standardised format b. Only the historical performance of the product c. Only the issuer's credit rating d. The full text of the prospectus

Answer and explanation

Answer: a. The highlights sheet is a short, standardised summary supporting comparison across products. It supplements rather than replaces the prospectus.

374. [**Multiple response**] Which statements about the product highlights sheet are correct?

a. It must be fair and balanced b. It should identify what the product is and its key risks c. It may omit the risks if the product is capital protected d. It supports comparison between similar products

Answer and explanation

Answer: a, b and d. A capital-protected product still carries issuer credit risk and liquidity risk, so risks cannot be omitted; that omission would also breach the fair-and-balanced requirement.

Appendix C — Futures contract specifications

375. [**Calculation**] A futures contract has a multiplier of SGD 50 and a tick size of 0.2 index points. What is the value of one tick?

a. SGD 0.20 b. SGD 10 c. SGD 50 d. SGD 100

Answer and explanation

Answer: b. Tick value

$$= \text{tick size} \times \text{multiplier} = 0.2 \times 50 = 10$$

.

376. [**Calculation**] A futures contract with a multiplier of SGD 25 moves from 4,000 to 4,008. What is the gain on one long contract?

a. SGD 8 b. SGD 200 c. SGD 800 d. SGD 2,000

Answer and explanation

Answer: b.

$$(4{,}008 - 4{,}000) \times 25 = 200$$

.

Appendix E — Review Questions

377. [**Single choice**] A review question asks which statements are true and then offers four lettered combinations. The question is testing:

a. Recall of a single definition b. Whether each statement can be evaluated independently and the results then combined c. The ability to compute a price d. Knowledge of the examiner's identity

Answer and explanation

Answer: b. Combination questions reward evaluating each statement separately, then choosing the option that matches exactly the set of true statements.

378. [**Multiple response**] Which exam techniques help with combination questions?

a. Mark each statement true or false before reading the options b. Eliminate options that include a statement you know to be false c. Look for two options that differ by only one statement and decide that statement d. Choose the longest option because it is usually correct

Answer and explanation

Answer: a, b and c. Statement-by-statement marking and elimination are sound. Option length carries no information, and relying on it is a guessing heuristic.

Appendix F — Essential readings

379. [**Single choice**] The essential readings listed for CM-SIP are best used:

a. As a substitute for the study guide b. As the primary source whenever the guide summarises a detail c. Only after the examination has been sat d. As a list to memorise without reading the underlying documents

Answer and explanation

Answer: b. The readings are the primary sources; the guide summarises them, so they are where ambiguous or examinable detail should be resolved.

380. [**Multiple response**] Which are listed as essential readings for CM-SIP?

a. The Securities and Futures Act b. The Code on Collective Investment Schemes c. The MAS product highlights sheet requirements for structured products d. The internal trading manual of a single bank

Answer and explanation

Answer: a, b and c. The readings are public legislation, MAS notices and industry codes. One firm's internal manual is not an essential reading.

Appendix G — Glossary and cross-reference

381. [**Single choice**] Applied to a call option, "in the money" means:

a. The underlying exceeds the strike b. The underlying is below the strike c. The option has already expired d. The premium exceeds the intrinsic value

Answer and explanation

Answer: a. A call is in the money when the underlying is above the strike, so exercise would produce a positive payoff.

382. [**Single choice**] "Contango" describes a futures market in which:

a. Futures prices fall as maturity lengthens b. Futures prices rise as maturity lengthens c. Spot exceeds futures at every maturity d. All contract months trade at the same price

Answer and explanation

Answer: b. Contango is an upward-sloping curve; backwardation is the reverse, with spot above futures.

383. [**Single choice**] A "covered call" involves:

a. Buying a call and holding cash b. Holding the underlying and selling a call against it c. Selling a put and buying the underlying d. Buying a put and selling a call

Answer and explanation

Answer: b. The long underlying covers the short call, capping upside in exchange for the premium received.

384. [**Single choice**] The "notional amount" of a derivative is:

a. The market value of the contract b. The reference amount on which payments are calculated c. The premium paid for the contract d. The margin posted with the clearing house

Answer and explanation

Answer: b. The notional is a reference principal used to scale payments. It is usually not exchanged, and it is not the contract's value.

385. [**Single choice**] A "mandatory call event" for a CBBC occurs when:

a. The issuer elects to redeem early b. The underlying touches the call price c. The investor fails to meet a margin call d. The contract reaches its expiry date

Answer and explanation

Answer: b. The event is triggered by the underlying reaching the call price, after which the contract terminates.

386. [**Single choice**] "Delta hedging" a short option position involves:

a. Trading the underlying to offset the option's delta exposure b. Buying an identical option to close the position c. Holding the position untouched until expiry d. Selling the underlying without regard to the option's sensitivity

Answer and explanation

Answer: a. Delta hedging offsets the first-order sensitivity of the option position by trading the underlying in the quantity the delta implies.

387. [**Single choice**] A "participation rate" in a structured product expresses:

a. The share of the underlying's upside passed to the investor b. The issuer's underwriting fee c. The probability that the issuer defaults d. The margin required against the position

Answer and explanation

Answer: a. It is the proportion of an index or asset gain the note delivers, with the remainder funding the structure's protection or margin.

388. [**Single choice**] A "backwardated" futures market is one in which:

a. Futures prices fall as maturity lengthens b. Futures prices rise as maturity lengthens c. The underlying is impossible to deliver d. All contract months trade at one price

Answer and explanation

Answer: a. Backwardation is a downward-sloping curve, typically reflecting a convenience yield that exceeds the carry cost.